
Linear Algebra I-II

Theorems & Lemmas

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1 Vector Spaces

1.1 Vector Spaces and Linear Subspaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V , endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (v_1, v_2) &\mapsto v_1 + v_2, \\ \cdot : K \times V &\rightarrow V, & (\alpha, v) &\mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

- (V1) $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$,
- (V2) $\exists$ an element $0 \in V : 0 + v = v \quad \forall v \in V$,
- (V3) $\forall v \in V \exists v' \in V : v + v' = 0$,
- (V4) $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1$,
- (V5) $\forall a, b \in K \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v$,
- (V6) $\forall v \in V : 1 \cdot v = v$,
- (V7) $\forall a \in K \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$,
- (V8) $\forall a, b \in K \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$.

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K . Then:

- (a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V .
- (b) The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,
- (c) $\forall v \in V : 0 \cdot v = 0$,
- (d) $\forall a \in K : a \cdot 0 = 0$,
- (e) $\forall v \in V : -1 \cdot v = -v$,
- (f) $\forall v \in V : -(-v) = v$,
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,
- (h) Associativity for $+$ holds also for some of n elements. $v_1 + \dots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\dots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\dots)$ etc.

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K .

Definition 1.3: Linear Subspace

Let V be a vector space over K . A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V .

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K , and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S . Define

$$K^S = \{f : S \rightarrow K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S$, $a \in K$. Define

$$\begin{aligned}(f + g) : S &\rightarrow K, & S \ni x &\mapsto f(x) + g(x), \\ (a \cdot f) : S &\rightarrow K, & S \ni x &\mapsto a \cdot f(x).\end{aligned}$$

Now Let $S = [0, 1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0, 1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

1.2 Span of Sets

Definition 1.6: Span

Let $S \subseteq V$ be a subset. We define

$$\text{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces $W \subseteq V$ that contain S . $\text{Sp}(S)$ is called the span of S (in V).

Lemma 1.7: Properties of the Span

- (1) $\text{Sp}(S) \subseteq V$ is a linear subspace.
- (2) If $W \subseteq V$ is a linear subspace and $S \subseteq W$, then $\text{Sp}(S) \subseteq W$.

In other words, $\text{Sp}(S)$ is a linear subspace of V , and moreover among those subspaces that contain S , $\text{Sp}(S)$ is the smallest one.

Definition 1.8: Linear Combination

Let $n \in \mathbb{N}$, $a_1, \dots, a_n \in K$, $v_1, \dots, v_n \in V$. A **linear combination** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

Lemma 1.9: Alternate Definition of Span

Let $\emptyset \neq S \subseteq V$. Then

$$\text{Sp}(S) = \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

Remark 1.10. The set S can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from S .

Let $n \in \mathbb{N}$, $v_1, \dots, v_n \in V$. We'll write

$$\text{Sp}(v_1, \dots, v_n) := \text{Sp}(\{v_1, \dots, v_n\}).$$

Remark 1.11. *It obviously holds that*

$$\text{Sp}(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \alpha_i v_i \mid \alpha_i \in K \quad \forall 1 \leq i \leq n \right\}.$$

Remark 1.12. *It follows from the definition of span that*

$$\text{Sp}(\emptyset) = \{0_V\}.$$

Definition 1.13: Generating Set

Let V be a vect. sp. over K and $S \subseteq V$. We say that S **generates** (or **spans**) V if $\text{Sp}(S) = V$. S is called a generating set of V if $\text{Sp}(S) = W$, where $W \subseteq V$ is a subspace of V . We say that S **generates** (or **spans**) W .

Definition 1.14: Finite Dimensional

A vector space V is called **finite-dimensional** if $\exists$ a **finite** set $S \subseteq V$ such that $\text{Sp}(S) = V$. Otherwise V is called **infinite-dimensional**.

1.3 Bases of Vector Spaces

1.3.1 Linear Independence

Let V be a vector space over K .

Definition 1.15: Linear Independence

Let $v_1, \dots, v_n$ be a list of n vectors in V . We say that $v_1, \dots, v_n$ is **linearly independent** if the only linear combination of $v_1, \dots, v_n$ that gives $0 \in V$ is the trivial one. In other words,

$$\forall a_1, \dots, a_n \in K : a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = 0, \dots, a_n = 0.$$

By definition, we say that the empty list of vectors $\emptyset$ is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

Remark 1.16. (1) 0 can always be written as a linear combination of any finite list of vectors $0v_1 + \dots + 0v_n = 0$, i.e., the trivial linear combination.

(2) If the list $v_1, \dots, v_n$ has two vectors that are the same, say $v_k = v_l$ for $1 \leq k < l \leq n$, then $v_1, \dots, v_n$ is **not** linearly independent. Indeed, $v_k + (-1)v_l = 0$.

(3) The order of the vectors in the list $v_1, \dots, v_n$ has no effect on linear independence. So, if there are no repetitions in $v_1, \dots, v_n$ we can talk about linear independence of the set $\{v_1, \dots, v_n\}$.

Definition 1.17: Linear Independence in the Context of Families

Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a family of vectors in V , indexed by a set I . We say that $\mathcal{F}$ is linearly independent if $\forall n \in \mathbb{N}$ and any sequence of distinct indices $i_1, \dots, i_n \in I$, the list of vectors $v_{i_1}, \dots, v_{i_n}$ is linearly independent. Of course, if in $\mathcal{F}$ there are any repetitions of vectors, i.e., if $\exists i, j, i \neq j$, s.t. $v_i = v_j$, then $\mathcal{F}$ cannot be linearly independent.

Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent

If $\emptyset \neq S \subseteq V$ is linearly independent. Then every non-empty subset $S' \subseteq S$ is also linearly independent.

Definition 1.19: Linear Dependence

A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is not linearly independent.

1.3.2 Basis of a Vector Space**Definition 1.20: Basis of a Vector Space**

A subset $S \subseteq V$ of a vector space is called a basis of V if the following holds:

- (1) S is linearly independent.
- (2) S generates (or spans) V , i.e., $\text{Sp}(S) = V$.

Proposition 1.21: Equivalent Formulation of basis

A subset $S \subseteq V$ is a basis of V if and only if every $v \in V$ can be written in a unique way as a linear combination of vectors in S .

Proof. For simplicity we will assume that $|S| < \infty$. The case $|S| = \infty$ is similar.

$\Leftarrow$: We assume that every $v \in V$ can be written in a unique way as a linear combination of elements from S . Now, "can be written" means that $\text{Sp}(S) = V$. Next write $S = \{v_1, \dots, v_n\}$. Consider the vector $0 \in V$. Clearly

$$0 = 0 \cdot v_1 + \dots + 0 \cdot v_n.$$

Since 0 can be written in a unique way as a linear combination of $v_1, \dots, v_n$ it follows that

$$0 = a_1 v_1 + \dots + a_n v_n \quad \implies \quad a_1 = \dots = a_n = 0.$$

Therefore, S is linearly independent, which proves that S is a basis of V .

$\Rightarrow$: We assume that S is a basis for V . By assumption $\text{Sp}(S) = V$, so this means every $v \in V$ can be expressed as a linear combination of vectors from S . It remains to show that this unique.

Assume by contradiction that $\exists v \in V$ which has two different ways to be written as a linear combination of vectors from S , i.e.,

$$v = a_1 v_1 + \dots + a_n v_n = b_1 v_1 + \dots + b_n v_n$$

and $a_k \neq b_k$ for some $1 \leq k \leq n$. Then

$$0 = (a_1 - b_1)v_1 + \dots + \underbrace{(a_k - b_k)}_{\neq 0} + \dots + (a_n - b_n)v_n$$

and we deduce that 0 can be written as a non-trivial linear combination of elements from S . Therefore, S is linearly dependent, which is a contradiction to the fact that S is a basis of V . $\square$

1.3.3 Example: Bases

$V = M_{m \times n}(K)$, $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis, where

$$E_{ij} = \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix},$$

where the 1 is at the position (i, j) in the matrix.

Lemma 1.22: A

Let $v_1, \dots, v_m \in V$ be a list of vectors which is linearly dependent. Then $\exists 1 \leq j \leq m$ s.t.:

- (1) $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$.
- (2) $\text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) = \text{Sp}(v_1, \dots, v_m)$. In other words, we can drop one of the vectors from the list without changing the span.

Lemma 1.23: B

Same assumptions as in Lemma 1.22, but we assume in addition that $v_1, \dots, v_k$ are linearly independent, where $1 \leq k \leq m$ is some index. Then $k < j$.

Lemma 1.24: C

Let $w_1, \dots, w_n \in V$ s.t. $\text{Sp}(w_1, \dots, w_n) = V$, and let $v \in V$. Then $v, w_1, \dots, w_n$ is linearly dependent.

Lemma 1.25: D

Let $v_1, \dots, v_n \in V$ be s.t. $\text{Sp}(v_1, \dots, v_n) = V$, and let $u_1, \dots, u_m \in V$ be linearly independent. Then $m \leq n$. Moreover, one can delete m vectors from the list $v_1, \dots, v_n$ s.t. the remaining list, say $v'_1, \dots, v'_{n-m}$ together with $u_1, \dots, u_m$ still span V , i.e., $\text{Sp}(u_1, \dots, u_m, v'_1, \dots, v'_{n-m}) = V$.

Lemma 1.26: E

Let $w_1, \dots, w_l \in V$ and assume that $w_j \notin \text{Sp}(w_1, \dots, w_{j-1}) \quad \forall 1 \leq j \leq l$. Then the list $w_1, \dots, w_l$ is linearly independent.

Lemma 1.27: F

Suppose that $\text{Sp}(v_1, \dots, v_n) = V$. Then $\exists$ a subset of $\{v_1, \dots, v_n\}$ which is a basis for V .

Theorem 1.28: Bases Have the Same Number of Elements

Let V be a finite dimensional vector space over K . Then V has a basis with finitely many elements. Moreover, every basis of V is finite and has the same number of elements.

Definition 1.29: Dimension of a Vector Space

Let V be a finite dimensional vector space over K . We define the dimension of V to be the number $n \in \mathbb{N}$ of elements in a basis (any basis !) of V . We write $\dim V := n$. In case V is not finite dimensional we write $\dim V = \infty$.

Remark 1.30. • $\dim V$ is well defined as it does not depend on a specific choice of basis for V .

- $\dim\{0\} = 0$, b.c. $\emptyset$ is a basis for the space $\{0\}$, and $|\emptyset| = 0$.

Summary

Let V be a finite dimensional vector space over K . Then:

- (1) Every finite list of vectors that spans V contains a sublist which is a basis for V .
- (2) Every linearly independent list can be extended to a basis of V .

Theorem 1.31: Equivalent Statements for Bases

Let V be a vector space of $\dim V = n \in \mathbb{N}$. Let $v_1, \dots, v_n \in V$. Then the following statements are equivalent:

- (1) $v_1, \dots, v_n$ are linearly independent.
- (2) $v_1, \dots, v_n$ generate V , i.e., $\text{Sp}(v_1, \dots, v_n) = V$.
- (3) $v_1, \dots, v_n$ is a basis for V .

Theorem 1.32: Consequences of Bases

Suppose that $n = \dim V < \infty$. Let $v_1, \dots, v_k \in V$. Then:

- (1) If $k < n$, the list $v_1, \dots, v_k$ **cannot** span V .
- (2) If $k > n$, the list $v_1, \dots, v_k$ **cannot** be linearly independent.

Proposition 1.33: Dimension of Subspaces

Let V be a finite dimensional vector space over K . Then every subspace $U \subseteq V$ is also finite dimensional. Moreover, $\dim U \leq \dim V$ with equality if and only if $U = V$.

1.4 Row and Column Spaces

Row and Column Spaces

Definition 1.34: Row and Column Space

Let $A \in M_{m \times n}(K)$ and $u_1, \dots, u_m \in K^n$ the rows of A and $v_1, \dots, v_n \in K^m$ the columns of A , i.e.,

$$A = \begin{pmatrix} - & u_1 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We define two spaces

$$\text{RowS}(A) := \text{Sp}(u_1, \dots, u_m) \subseteq K^n$$

$$\text{ColS}(A) := \text{Sp}(v_1, \dots, v_n) \subseteq K^m.$$

Lemma 1.35: Row Equivalent Matrices

If $A, B \in M_{m \times n}(K)$ are row-equivalent matrices then $\text{RowS}(A) = \text{RowS}(B)$.

Remark 1.36. The statement of the Lemma does **not** hold for the column space.

1.4.1 Transpose of a Matrix

Definition 1.37: Transpose of a Matrix

Let $A = (a_{i,j}) \in M_{m \times n}(K)$. We define the transposed matrix of A as $A^T := (b_{i,j}) \in M_{n \times m}(K)$, where $b_{i,j} = a_{j,i}$.

Lemma 1.38: Properties of the Transpose

or every $A, B \in M_{m \times n}(K)$, $\lambda \in K$ we have

1. $(A + B)^T = A^T + B^T$.
2. $(\lambda A)^T = \lambda A^T$.
3. $(A^T)^T = A$.
4. $\text{RowS}(A^T) = \text{ColS}(A)$, $\text{ColS}(A^T) = \text{RowS}(A)$.

1.5 Sums of Vector Spaces

Definition 1.39: Sum of Vector Spaces

Let V be a vector space over K and $U, W \subseteq V$ linear subspaces. We define the sum $U + W$ of U, W as

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way we can define sums of more than two subspaces. If $U_1, \dots, U_r \subseteq V$ are subspaces, define

$$U_1 + \dots + U_r := \{u_1 + \dots + u_r \mid u_1 \in U_1, \dots, u_r \in U_r\}.$$

Proposition 1.40: Properties of the Sum of Vector Spaces

Let $U, W \subseteq V$ be linear subspaces. Then:

1. $U + W = \text{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is a subspace and $U, W \subseteq U + W$ are subspaces of $U + W$.
2. If U, W are finite dimensional then so is $U + W$. Moreover, one can find a basis for $U + W$ as follows:
 - (a) Choose a basis $p_1, \dots, p_k$ for $U \cap W$, where $k = \dim(U \cap W)$.
 - (b) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, u_1, \dots, u_{l-k}$ of U , where $l = \dim U$.
 - (c) Extend $p_1, \dots, p_k$ to a basis $p_1, \dots, p_k, w_1, \dots, w_{m-k}$ of W , where $m = \dim W$.

Then $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$ is a basis for $U + W$.

3. In particular, we have $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Corollary 1.41: Consequences of the Properties

Let $U, W \subseteq V$ be two finite dimensional linear subspaces of a vector space V . The following are equivalent:

1. $\dim(U + W) = \dim U + \dim W$.
2. $\dim(U \cap W) = 0$.
3. $U \cap W = \{0\}$.
4. Every $v \in U + W$ can be written in a **unique** way as $v = u + w$ with $u \in U, w \in W$.
5. If $u + w = 0$, where $u \in U$ and $w \in W$, then $u = w = 0$.

Definition 1.42: Linear Complement

Let V be a vector space over K and $U \subseteq V$ a linear subspace. A linear subspace $W \subseteq V$ is called a **linear complement** of U if $U + W = V$ and $U \cap W = \{0\}$. Note that U, W satisfy all the five equivalent statements from the corollary 1.41.

Proposition 1.43: Existence of a Complement

Let V be a finite dimensional vector space and $U \subseteq V$ a subspace. Then there exists a subspace $W \subseteq V$ which is a complement to U .

2 Linear Maps

Definition 2.1: Linear Map

Let V, W be vector spaces over K . A map $T : V \rightarrow W$ is called linear map if:

1. $\forall v_1, v_2 \in V : T(v_1 + v_2) = T(v_1) + T(v_2)$
2. $\forall \alpha \in K, \forall v \in V : T(\alpha \cdot v) = \alpha \cdot T(v)$

This means that $T : V \rightarrow W$ is a map that respects the structures of vector spaces on $(V, +, \cdot), (W, +, \cdot)$. We denote the set of all linear maps $T : V \rightarrow W$ by $\text{Hom}(V, W)$. Sometimes $\text{hom}_K(V, W)$.

Proposition 2.2: Properties of Linear Maps

Let V, W be vector spaces over a field K , and let $T : V \rightarrow W$ be a linear map. Then:

1. $\forall n \in \mathbb{N}, a_1, \dots, a_n \in K, v_1, \dots, v_n \in V$ we have

$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

2. $T(0_V) = 0_W$.

Theorem 2.3: A unique Linear Map

Let V be a finite dimensional vector space and $v_1, \dots, v_n \in V$ a basis for V . Let $w_1, \dots, w_n \in W$ be arbitrary vectors. Then there exists a unique linear map $T : V \rightarrow W$ s.t.

$$T(v_i) = w_i \quad \forall 1 \leq i \leq n.$$

Lemma 2.4: Matrix Representation of the Unique Linear Map

For every linear transformation $T : K^n \rightarrow K^m$ there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $T = T_A$.

2.1 Kernel & Image

Definition 2.5: Kernel and Image

Let V, W be vector spaces over K and $T : V \rightarrow W$ a linear map.

1. The set

$$\ker(T) := \{v \in V \mid Tv = 0\} \subseteq V$$

is a linear subspace of V . It is called the **kernel of T** .

2. The set

$$\text{Im}(T) := \{Tv \mid v \in V\} \subseteq W$$

is a linear subspace of W . It is called the **image of T** .

With the notation from set theory, we have

$$\ker(T) = T^{-1}(\{0\}), \quad \text{Im}(T) = T(V).$$

Proposition 2.6: Equivalences for Injective and Surjective

Let $T : V \rightarrow W$ be a linear map. Then:

1. T is injective $\iff \ker(T) = \{0\}$.
2. T is surjective $\iff \text{Im}(T) = W$.

2.1.1 About Linear Maps

Definition 2.7: Isomorphism

A linear map $T : V \rightarrow W$ is called an **isomorphism** if there exists a linear map $S : W \rightarrow V$ s.t. $S \circ T = \text{id}_V$, $T \circ S = \text{id}_W$. We say that V is isomorphic to W if there exists an isomorphism $T : V \rightarrow W$. We write $V \cong W$.

Lemma 2.8: Linearity of the Inverse Map

Let $T : V \rightarrow W$ be a linear map which is bijective. Then the inverse map $T^{-1} : W \rightarrow V$ is linear. In other words, a linear bijective map is an isomorphism.

Lemma 2.9: Composition of Linear Maps

Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear maps. Then the composition $S \circ T : V \rightarrow U$ is also linear.

Definition 2.10: Endomorphism / Automorphism

A linear map $T : V \rightarrow V$ is sometimes called an **endomorphism** of V . We denote the set of all endomorphisms of V by $\text{End}(V)$. (So, $\text{End}(V) = \text{Hom}(V, V)$.)

A linear map $T : V \rightarrow V$ which is an isomorphism is also called an **automorphism** of V .

Lemma 2.11: Span of a Basis

Let $T : V \rightarrow W$ be a linear map. Let $v_1, \dots, v_n$ be a basis of V . Then $\text{Im}(T) = \text{Sp}(Tv_1, \dots, Tv_n)$

Theorem 2.12: Rank Theorem

Let V, W be vector spaces, where V is finite dimensional and let $T : V \rightarrow W$ be a linear map. Then

$$\dim \ker(T) + \dim \text{Im}(T) = \dim V.$$

Proof. Define $n := \dim V$. Let $u_1, \dots, u_k$ be a basis for $\ker(T)$. Note that $\ker(T)$ is finite dimensional, b.c. it is a subspace of V , which is finite dimensional. Extend $u_1, \dots, u_k$ to a basis $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ of V , which we can do by a previous result. By Lemma 2.11, we have that

$$\text{Im}(T) = \text{Sp}(\underbrace{T u_1, \dots, T u_k}_{=0}, T v_1, \dots, T v_{n-k}) = \text{Sp}(T v_1, \dots, T v_{n-k}).$$

We'll show now that $T v_1, \dots, T v_{n-k}$ is linearly independent. To see this, assume that

$$\sum_{i=1}^{n-k} a_i T v_i = 0,$$

for some $a_1, \dots, a_{n-k} \in K$. Since T is linear we can write this linear combination as

$$T\left(\sum_{i=1}^{n-k} a_i v_i\right) = 0 \quad \implies \quad \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

Now, $u_1, \dots, u_k$ form a basis for $\ker(T)$. So, $\exists b_1, \dots, b_k \in K$ s.t.

$$\begin{aligned} \sum_{i=1}^{n-k} a_i v_i &= b_1 u_1 + \dots + b_k u_k \\ \implies b_1 u_1 + \dots + b_k u_k + (-a_1) v_1 + \dots + (-a_{n-k}) v_{n-k} &= 0. \end{aligned}$$

Since, $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ form a basis for V , we have that $b_1 = \dots = b_k = a_1 = \dots = a_{n-k} = 0$. This shows that $T v_1, \dots, T v_{n-k}$ are linearly independent. Therefore, $T v_1, \dots, T v_{n-k}$ form a basis for $\text{Im}(T)$. Thus,

$$\begin{aligned} \dim \text{Im}(T) &= n - k = \dim V - \dim \ker(T) \\ \implies \dim \ker(T) + \dim \text{Im}(T) &= \dim V. \end{aligned}$$

□

Corollary 2.13: Consequences of the Rank Theorem

Let $T : V \rightarrow W$ be a linear map, where V, W are finite dimensional vector spaces. Then the following holds:

1. If $\dim W < \dim V$ then T is **not** injective.
2. If $\dim W > \dim V$ then T is **not** surjective.
3. If $\dim W = \dim V$ then T is bijective $\iff T$ is surjective $\iff T$ is injective.

Corollary 2.14: Equivalence for an Isomorphism

Two finite dimensional vector spaces V, W are isomorphic if and only if $\dim V = \dim W$.

Theorem 2.15: Image of a Set

Let $T : V \rightarrow W$ be an isomorphism between two vector spaces V, W . Let S be a family of vector in V . Write $T(S) := \{Tv\}_{v \in S}$. Then:

1. S is linearly independent if and only if $T(S)$ is linearly independent.
2. S spans V if and only if $T(S)$ spans W .
3. S is a basis of V if and only if $T(S)$ is a basis of W .

What is the significance of isomorphisms? If two vector spaces V, W are isomorphic, $V \cong W$ (i.e., $\exists T : V \rightarrow W$ bijection), then V and W have the same linear algebraic properties (like dimension). We can think of V and W as the same space represented in two different ways.

However, identifying V with W usually requires a choice of an isomorphism $T : V \rightarrow W$. Usually there is no preferred (canonical) choice. It is therefore better, in general, not to identify different vector spaces that are isomorphic.

Definition 2.16: Rank

Let $T : V \rightarrow W$ be a linear map. We define the **rank** of T by

$$\text{rank}(T) := \dim \text{Im}(T)$$

Lemma 2.17: Rank of the Composition

Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps, where U, V, W are finite dimensional vector spaces. Then

1. $\text{rank}(S \circ T) \leq \min\{\text{rank}(S), \text{rank}(T)\}$.
2. If S is injective, then $\text{rank}(S \circ T) = \text{rank}(T)$.
3. If T is surjective, then $\text{rank}(S \circ T) = \text{rank}(S)$.

2.2 Writing Linear Maps using Bases/Coordinates

In this Section we'll assume all vector spaces to be finite dimensional unless otherwise stated. We'll write bases of a vector space V as an ordered list $\mathcal{B} = (v_1, \dots, v_n)$.

Definition 2.18: Coordinate Map

Let V be a vector space over K and $\mathcal{B} = (v_1, \dots, v_n)$ a basis of V . Let $v \in V$. We define the coordinate-vector of v w.r.t. the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique scalars in K s.t. $v = a_1 v_1 + \dots + a_n v_n$. Note that $[v]_{\mathcal{B}}$ is uniquely defined since every $v \in V$ has a unique representation as a linear combination of $v_1, \dots, v_n$.

Define a map $\Phi_{\mathcal{B}} : V \rightarrow K^n$ as

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

Proposition 2.19: $\Phi_{\mathcal{B}}$ is an Isomorphism

$\Phi_{\mathcal{B}}$ is an isomorphism

Remark 2.20. The inverse map to $\Phi_{\mathcal{B}}$ is given by

$$\Phi_{\mathcal{B}}^{-1} : K^n \rightarrow V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \dots + a_n v_n.$$

Corollary 2.21: Isomorphism Coordinate Space

Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then $V \cong K^n$.

Proof. Choose a basis $\mathcal{B}$ for V . Then $\Phi_{\mathcal{B}} : V \rightarrow K^n$ is an isomorphism. □

Remark 2.22. 1. The corollary does not hold for $\dim V = \infty$, at least not as stated.

2. We've seen that $V \cong K^n$, if V is finite dimensional, but in general there does **not** exist a canonical isomorphism $V \xrightarrow{\cong} K^n$. For example $\Phi_{\mathcal{B}}$ depends on the choice of basis $\mathcal{B}$.

Now we arrive at the main question. Let V, W be vector spaces over K , $n := \dim V, m := \dim W$, and let $T : V \rightarrow W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W . Consider the diagram

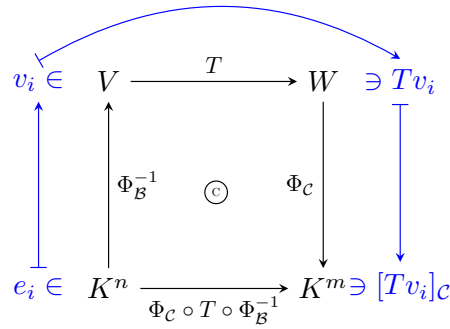
$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \Phi_{\mathcal{B}}^{-1} \uparrow & \text{\textcircled{C}} & \downarrow \Phi_{\mathcal{C}} \\ K^n & \xrightarrow{\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}} & K^m \end{array}$$

The 'C' means that the diagram is commutative.

Since $\Phi_{\mathcal{B}}^{-1}, T, \Phi_{\mathcal{C}}$ are all linear maps, the map $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} : K^n \rightarrow K^m$ is also linear. By Lemma 2.4, any linear map $K^n \rightarrow K^m$ can be written using a matrix $A \in M_{m \times n}(K)$. In other words, there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} = T_A$. The matrix A is called the **representation matrix of T w.r.t. the bases $\mathcal{B}$ and $\mathcal{C}$** . We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K)$.

2.2.1 Entries of the Representation Matrix

Let $T : V \rightarrow W$ be a linear map and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V , and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W .



$T_A(e_i) = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}(e_i) = \Phi_{\mathcal{C}} \circ T v_i = [T v_i]_{\mathcal{C}}$. But these are precisely the columns of A . In other words

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T v_1]_{\mathcal{C}} & \dots & [T v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in M_{m \times n}.$$

Another way to describe this matrix is: Let $A = (a_{i,j})$, where $a_{i,j}$ are defined uniquely by

$$T v_j = \sum_{i=1}^m a_{i,j} w_i.$$

Proposition 2.23: Representation matrix

Let $T : V \rightarrow W$ be a linear map between two finite dimensional vector spaces V, W . Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then

$$\forall v \in V : [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T v]_{\mathcal{C}}.$$

2.2.2 Representing Composition of Linear Maps by Matrices

Let $T : V \rightarrow W, S : W \rightarrow U$ be linear maps. Let $\mathcal{A} = (v_1, \dots, v_n)$ be a basis for V , $\mathcal{B} = (w_1, \dots, w_m)$ a basis for W and $\mathcal{C} = (u_1, \dots, u_p)$ a basis for U .

Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})$, $[S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij})$, $[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij})$. By definition we have that

$$T(v_j) = \sum_{k=1}^m a_{kj} w_k \quad \forall 1 \leq j \leq n \quad (2.1)$$

$$T(w_j) = \sum_{i=1}^p b_{ij} u_i \quad \forall 1 \leq j \leq m \quad (2.2)$$

$$S \circ T(v_j) = \sum_{i=1}^p c_{ij} u_i \quad \forall 1 \leq j \leq n \quad (2.3)$$

But we can calculate $S \circ T(v_j)$ by using Equations (2.1), (2.2). Then we have

$$\begin{aligned} S \circ T(v_j) &= S\left(\sum_{k=1}^m a_{kj} w_k\right) = \sum_{k=1}^m a_{kj} S(w_k) = \sum_{k=1}^m a_{kj} \left(\sum_{i=1}^p b_{ik} u_i\right) \\ &= \sum_{k=1}^m \sum_{i=1}^p a_{kj} b_{ik} u_i = \sum_{i=1}^p \sum_{k=1}^m a_{kj} b_{ik} u_i = \sum_{i=1}^p \left(\sum_{k=1}^m b_{ik} a_{kj}\right) u_i \\ \implies c_{ij} &= \sum_{k=1}^m b_{ik} a_{kj}. \end{aligned}$$

3 Matrices

Definition 3.1: Matrix Multiplication

Let $A \in M_{m \times n}(K)$, $B \in M_{n \times p}(K)$. We define a new matrix $A \cdot B \in M_{m \times p}(K)$ which is called the product of A and B . Write $A = (a_{ij})$, $B = (b_{ij})$. The matrix $A \cdot B$ has entries (c_{ij}) , where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Important: $A \cdot B$ is defined only when the number of columns of A equals the number of rows of B .

Conclusion: Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps between finite dimensional vector spaces. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be bases for V, W, U respectively. Then

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = [S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}}.$$

Definition 3.2: Kronecker-Delta/Identity Matrix

Let $\mathcal{I}$ be an index set. Define for all $i, j \in \mathcal{I}$

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

This is called the **kronecker-delta**.

We define the **identity matrix** $I_n \in M_{n \times n}(K)$ as

$$I_n := \delta_{ij}, \quad I_n = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Remark 3.3. Let V be a finite dimensional vector space and $n := \dim V$. Let $\text{id} : V \rightarrow V$ be the identity map. Let $\mathcal{B}$ be any basis for V . Then $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = I_n$. In other words, the matrix representation of the identity map w.r.t $\mathcal{B}$ is the identity matrix I_n .

Proposition 3.4: Associativity and Distributivity of Matrix Multiplication

$$1. \forall A \in M_{m \times n}(K), B \in M_{n \times p}(K), C \in M_{p \times q}(K) : (AB)C = A(BC) \in M_{m \times q}.$$

$$2. \forall A, B \in M_{m \times n}(K), C \in M_{n \times p}(K), C' \in M_{q \times m}(K) \text{ we have}$$

$$(A + B)C = AC + BC \in M_{m \times p}(K),$$

$$C'(A + B) = C'A + C'B \in M_{q \times n}(K).$$

$$3. \forall A \in M_{m \times n}(K) : I_m A = A I_n = A.$$

$$4. \forall \alpha \in K, A \in M_{m \times n}(K), B \in M_{n \times p}(K) : \alpha(AB) = (\alpha A)B = A(\alpha B).$$

Remark 3.5. Matrix multiplication is in general **not** commutative, i.e., $\exists A, B$ s.t. $AB \neq BA$.

In $M_{n \times n}(K)$ we can add/subtract matrices and also multiply matrices. There is also a unit I_n . These operations satisfy properties like associativity, distributivity, etc. From this we get that $M_{n \times n}(K)$ is a (non-commutative) **ring** with a unity.

Definition 3.6: Invertibility

An $n \times n$ matrix $A \in M_{n \times n}(K)$ is **invertible** if $\exists B \in M_{n \times n}(K)$ s.t. $AB = BA = I_n$.

Remark 3.7. If A is invertible, then there exists only one matrix B s.t. $AB = BA = I_n$.

Proof. Assume $AB = BA = I_n$ and $AC = CA = I_n$. Then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

□

Definition 3.8: The Inverse Matrix

If A is invertible, the unique matrix B s.t. $AB = BA = I_n$ is called **the inverse of A** and is denoted by A^{-1} . So, we have $AA^{-1} = A^{-1}A = I_n$.

We denote the set of all invertible $n \times n$ matrices by

$$GL_n(K) = GL(n, K) := \{A \in M_{n \times n}(K) \mid A \text{ is invertible}\}.$$

This set is called the general linear group.

Proposition 3.9: Structure of the General Linear Group

$GL_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$ is a group. The unity is I_n . The inverse of $A \in GL_n(K)$ is A^{-1} . Moreover,

$$\forall A, B \in GL_n(K) : (A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

Corollary 3.10: Unique Solution for Linear System of Equations

Let $A \in GL_n(K)$. Then $\forall b \in K^n$ the linear system of equations $Ax = b$ has a unique solution, namely $x = A^{-1}b$.

Proposition 3.11: Invertibility and Isomorphisms

Let $A \in M_{n \times n}(K)$. Then $A \in GL_n(K)$ if and only if $T_A : K^n \rightarrow K^n$ is an isomorphism. Moreover, $(T_A)^{-1} = T_{A^{-1}}$.

Definition 3.12: Upper/Lower Triangular and Diagonal Matrices

A matrix $A \in M_{n \times n}(K)$, $A = (a_{ij})$, is called

- **Upper triangular** if $a_{ij} = 0 \quad \forall i > j$, i.e.,

$$\begin{pmatrix} * & \dots & \dots & * \\ 0 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

- **Lower Triangular** if $a_{ij} = 0 \quad \forall i < j$, i.e.,

$$\begin{pmatrix} * & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 0 \\ * & \dots & \dots & * \end{pmatrix}.$$

- **Diagonal** if $a_{ij} = 0 \quad \forall i \neq j$, i.e.,

$$\begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Lemma 3.13: Multiplication of Triangular/Diagonal Matrices

Let A, B be two diagonal (respectively both upper triangular, respectively both lower triangular). Then $A \cdot B$ is of the same type.

Corollary 3.14: Change of Bases for a Linear Map

Let V, W be finite dimensional vector spaces over K . Let $n := \dim V, m := \dim W$. Let $\mathcal{B}, \mathcal{B}'$ be two bases for V , and $\mathcal{C}, \mathcal{C}'$ be two bases for W . Then

$$[id_V]_{\mathcal{B}}^{\mathcal{B}'} \in GL_n(K), [id_W]_{\mathcal{C}'}^{\mathcal{C}} \in GL_m(K), \text{ and} \\ [id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}, [id_V]_{\mathcal{C}}^{\mathcal{C}'} = ([id_V]_{\mathcal{C}'}^{\mathcal{C}})^{-1}.$$

Let $T : V \rightarrow W$ be a linear map. Then

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [id_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

Definition 3.15: Transition Matrix/Change of Bases Matrix

Let V be a finite dimensional vector space over K , $n := \dim V$. Let $\mathcal{B}, \mathcal{B}'$ be two bases of V . The matrix $[id_V]_{\mathcal{B}'}^{\mathcal{B}} \in GL_n(K)$ is called the **change of bases matrix** between $\mathcal{B}$ and $\mathcal{B}'$, or the **transition matrix** from basis $\mathcal{B}$ to basis $\mathcal{B}'$.

If $\mathcal{B} = (v_1, \dots, v_n), \mathcal{B}' = (v'_1, \dots, v'_n)$ then

$$[id_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{B}'} & \dots & [v_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}.$$

Proposition 3.16: Inverse of a Matrix with Bases

Let $T : V \rightarrow W$ be an isomorphism between two finite dimensional vector spaces. Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then $[T]_{\mathcal{C}}^{\mathcal{B}}$ is an invertible matrix and

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}.$$

Corollary 3.17: Change of Basis for an Endomorphism

Let V be a finite dimensional vector space and $T : V \rightarrow V$ a linear map (an endomorphism). Let $\mathcal{B}, \mathcal{B}'$ be two bases for V . Then

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

Definition 3.18: Equivalence and Similarity of Matrices

1. Let $A, B \in M_{m \times n}(K)$. We say that A and B are **equivalent** if $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t. $B = PAQ$.
2. Let $A, B \in M_{n \times n}(K)$. We say that A and B are **similar** if $\exists P \in GL_n(K)$ s.t. $B = P^{-1}AP$.

Remark 3.19. 1. Equivalence between $m \times n$ matrices gives an equivalence relation on $M_{m \times n}(K)$.

2. $A, B \in M_{m \times n}(K)$ are equivalent if and only if there exists two vector spaces V, W with $n := \dim V$, $m := \dim W$ and two bases $\mathcal{B}, \mathcal{B}'$ of V and two bases $\mathcal{C}, \mathcal{C}'$ for W s.t. $A = [T]_{\mathcal{C}}^{\mathcal{B}}$, $B = [T]_{\mathcal{C}'}^{\mathcal{B}'}$. In other words, A is equivalent to B if and only if A and B are matrix representations of the same linear map w.r.t. different choices of bases for V and W .

3. Similarity between $n \times n$ matrices gives an equivalence relation on $M_{n \times n}(K)$.

4. $M, N \in M_{n \times n}(K)$ are similar if and only if there exists an n -dimensional vector space V and $T : V \rightarrow V$ and two bases $\mathcal{B}, \mathcal{B}'$ of V s.t. $M = [T]_{\mathcal{B}}^{\mathcal{B}}$, $N = [T]_{\mathcal{B}'}^{\mathcal{B}'}$.

Proposition 3.20: Existence of 'Nice' Bases

Let V, W be finite dimensional vector spaces over K , of dimension $n := \dim V$, $m := \dim W$. Let $T : V \rightarrow W$ be a linear map with $\text{rank}(T) = r$. (Recall that $\text{rank}(T) := \dim \text{Im}(T)$.) Then there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \dots, w_m)$ of W s.t.

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

Corollary 3.21

Let $A \in M_{m \times n}(K)$. Then $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix},$$

where $r = \text{col-rank}(A)$. In particular, $\text{col-rank}(A) \leq \min\{m, n\}$.

Corollary 3.22: Condition for Equivalence of Matrices

Let $A, B \in M_{m \times n}(K)$. Then $A \sim B$ if and only if $\text{col-rank}(A) = \text{col-rank}(B)$.

Moreover, $\forall 0 < r \leq \min\{m, n\}$ there is precisely one equivalence class of matrices A with $\text{col-rank} = r$.

3.1 col-rank = row-rank**Theorem 3.23: col-rank = row-rank**

Let $A \in M_{m \times n}(K)$. Then $\text{col-rank}(A) = \text{row-rank}(A)$.

Remark 3.24. 1. Because of this Theorem we'll denote from now on $\text{rank}(A) := \text{col-rank}(A) = \text{row-rank}(A)$.

2. Suppose A' is a row-reduced echelon matrix which is row-equivalent to A . Then we have that $\text{rank}(A) = \# \text{ pivots in } A'$. Indeed, $\text{rank}(A) = \text{row-rank}(A) = \text{row-rank}(A') = \# \text{ pivots in } A'$.
3. Recall that for $T : V \rightarrow W$ we defined $\text{rank}(T) := \dim \text{Im}(T)$. Also, if $A \in M_{m \times n}(K)$, then $\text{rank}(T_A) = \dim \text{Im}(T) = \text{col-rank}(A)$.

We now present some Lemmas for the proof of the Theorem.

Lemma 3.25: Isomorphisms preserve the Rank

Let $T : V \rightarrow W$ be a linear map, and $S : W \xrightarrow{\cong} U$, $L : P \xrightarrow{\cong} V$ be isomorphisms. Then

1. $\text{rank}(S \circ T) = \text{rank}(T)$.
2. $\text{rank}(T \circ S) = \text{rank}(T)$.

Lemma 3.26

Let $A \in M_{m \times n}(K)$. Then A is invertible if and only if $T_A K^n \rightarrow K^n$ is an isomorphism.

Lemma 3.27: Equivalent Matrices Have the Same Rank

Let $A, B \in M_{m \times n}(K)$ s.t. $B = PAQ$, where $P \in GL_m(K)$, $Q \in GL_n(K)$. Then

1. $\text{col-rank}(A) = \text{col-rank}(B)$
2. $\text{row-rank}(A) = \text{row-rank}(B)$.

Proof. Proof of Theorem 3.23: By Corollary 3.21, $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

Denote this matrix by B . For B we have $\text{col-rank}(B) = \text{row-rank}(B) = r$. By Lemma 3.27, we have that $\text{col-rank}(A) = \text{col-rank}(B) = r$, and $\text{row-rank}(A) = \text{row-rank}(B) = r$. Therefore, $\text{col-rank}(A) = \text{row-rank}(A)$. $\square$

Remark 3.28. Although $\text{row-rank}(A) = \text{col-rank}(A)$, the spaces $\text{ColS}(A)$ and $\text{RowS}(A)$ are **different**. First of all if $A \in M_{m \times n}$, then $\text{ColS}(A) \subseteq K^m$ and $\text{RowS}(A) \subseteq K^n$. But even when $n = m$, these two spaces are **different**. They have the same dimension, but are different.

3.2 More on Matrices and Linear Equations

Let $A \in M_{m \times n}(K)$, and consider the system of linear equations $Ax = 0$. Denote by

$$\text{Sol}(A, 0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A),$$

the **space** of solutions $Ax = 0$. Note that $\dim \text{Sol}(A, 0) = n - \text{rank}(A)$ (b.c. $\dim \ker(T_A) = n - \dim \text{Im}(T_A)$).

Let A' be a row-reduced echelon matrix which is row-equivalent to A (e.g. A' is obtained from A by Gauss-elimination). # pivots of $A' = \text{rank}(A)$, $n - \text{rank}(A) = \#$ free variables. Also, we have that $\text{Sol}(A, 0) = \text{Sol}(A', 0)$.

$$\begin{pmatrix} 0 & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Denote by $x_{i_1}, \dots, x_{i_l}$ the free variables, where $1 \leq i_1 < \dots < i_l \leq n$, $l = n - \text{rank}(A)$ and by $x_{j_1}, \dots, x_{j_k}$, where $1 \leq j_1 < \dots < j_k \leq n$, $r = \text{rank}(A)$ the pivots. For each choice of values for the free variables $x_{i_1}, \dots, x_{i_l} \in K$, the pivot variables $x_{j_1}, \dots, x_{j_k}$ are uniquely determined. So, we obtain a map

$$\Phi : K^l \rightarrow \text{Sol}(A, 0)$$

$$\Phi(\lambda_1, \dots, \lambda_l) = \text{the unique solution of } Ax = 0,$$

with $x_{i_1} = \lambda_1, \dots, x_{i_l} = \lambda_l$.

Proposition 3.29: Φ is an Isomorphism

The map Φ is linear. Moreover, it is an isomorphism.

Corollary 3.30

Let $b \in K^m$, $A \in M_{m \times n}(K)$. Then

1. $\text{Sol}(A, b) := \{x \in K^n \mid Ax = b\} \neq \emptyset$ iff $b \in \text{ColS}(A)$.
2. If $\text{Sol}(A, b) \neq \emptyset$ and $y \in \text{Sol}(A, b)$ then

$$\text{Sol}(A, b) = y + \text{Sol}(A, 0) := \{y + x \mid x \in \text{Sol}(A, 0)\}.$$

Proposition 3.31

Let $A \in M_{m \times n}(K)$, $b \in K^m$. The following statements are equivalent:

1. $\text{rank}(A|b) = \text{rank}(A)$.
2. The system of equations $Ax = b$ has a solution.

Proposition 3.32

1. $\text{rank}(A|b) = \text{rank}(A) = n$.
2. The system of equations $Ax = b$ has a unique solution.

3.3 Elementary Row Operations Revisited

Let $A \in M_{n \times p}(K)$. We learned about elementary row operations which are used as part of the Gauss elimination. All of these operations can be expressed by matrix multiplications.

Denote by E_{ij} the matrix in Example 1.3.3.

Type I: Let $1 \leq i \leq n$, $1 \leq j \leq n$, $i \neq j$. Let $\alpha \in K$. Define $Q_{ij}(\alpha)$ to be the matrix obtained from I_n after applying $R_i + \alpha R_j \rightarrow R_i$. We have $Q_{ij}(\alpha) = I_n + \alpha E_{ij}$.

Type II: Let $1 \leq i \leq n$, $1 \leq j \leq n$. Define P_{ij} to be the matrix obtained from I_n by permuting (exchanging) row $\# i$ and row $\# j$, i.e. applying $R_i \leftrightarrow R_j$. We have $P_{ij} = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$.

Type III: Let $1 \leq i \leq n$ and $0 \neq \alpha \in K$. Denote by $S_i(\alpha)$ the matrix obtained from I_n by applying $\alpha R_i \rightarrow R_i$.

Lemma 3.33: Elementary Row Operations as Matrix Multiplication

Let $A \in M_{n \times p}(K)$.

1. Multiplying A from the left with $Q_{ij}(\alpha)$ results in the row operation $R_i + \alpha R_j \rightarrow R_i$: $A \xrightarrow{R_i + \alpha R_j \rightarrow R_i} Q_{ij}(\alpha)A$.
2. Multiplying A from the left with P_{ij} results in the row operation $R_i \leftrightarrow R_j$: $A \xrightarrow{R_i \leftrightarrow R_j} P_{ij}A$.
3. Multiplying A from the left with $S_i(\alpha)$ results in the row operation $\alpha R_i \rightarrow R_i$: $A \xrightarrow{\alpha R_i \rightarrow R_i} S_i(\alpha)A$.

The same holds for elementary column operations, which is done by right hand side multiplications of the matrices.

Lemma 3.34: Elementary Matrices are Invertible

Every elementary matrix is invertible, and the inverse of each of them is an elementary matrix of the same type, i.e.

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \quad P_{ij}^{-1} = P_{ji}, \quad S_{ij}(\alpha)^{-1} = S_{ij}(\alpha^{-1}).$$

Theorem 3.35

$\forall A \in GL_n(K)$, $\exists k \geq 1$ and elementary matrices $T_1, \dots, T_k$ s.t. $T_k \cdot \dots \cdot T_1 \cdot A = I_n$. Furthermore, we have that

$$A = T_1^{-1} \cdot \dots \cdot T_k^{-1}, \quad A^{-1} = T_k \cdot \dots \cdot T_1.$$

In particular, A can be written as a finite product of elementary matrices.

Theorem 3.36: Equivalent condition for Invertibility

Let $A \in M_{n \times n}(K)$. Place A together with I_n in the following way $(A|I_n)$. Then A is invertible if and only if $(A|I_n)$ can be brought via a sequence of elementary row-operations to the form $(I_n|B)$ for some matrix $B \in M_{n \times n}(K)$. In this case we have $A^{-1} = B$.

4 Constructing New Vector Spaces Out of Old Ones

Proposition 4.1: The Vector Space of Linear Functions

Let V, W be two vector spaces over K . Then $\text{Hom}(V, W)$ has the structure of a vector space over K , if we endow it with the following operations:

- $\forall T_1, T_2 \in \text{Hom}(V, W)$, $(T_1 + T_2)(v) := T_1(v) + T_2(v) \quad \forall v \in V$.
- $\forall T \in \text{Hom}(V, W)$, $\alpha \in K$, $(\alpha T)(v) := \alpha T(v) \quad \forall v \in V$.

The neutral element of $\text{Hom}(V, W)$ is the map $0 : V \rightarrow W$, $v \mapsto 0$.

Theorem 4.2: The map $\Psi_C^{\mathcal{B}}$

Let V, W be finite dimensional vector spaces over K , $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V , and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W . Then the map

$$\Psi_C^{\mathcal{B}} : \text{Hom}(V, W) \rightarrow M_{m \times n}(K)$$

defined by

$$T \mapsto [T]_{\mathcal{C}}^{\mathcal{B}} := \begin{pmatrix} | & & | \\ [Tv_1]_{\mathcal{C}} & \dots & [Tv_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}$$

is a linear map and moreover an isomorphism.

Corollary 4.3: Dimension of $\text{Hom}(V, W)$

If V, W are finite dimensional vector spaces then so is $\text{Hom}(V, W)$ and $\dim \text{Hom}(V, W) = \dim V \cdot \dim W$.

4.0.1 Rings

Recall that Rings have the same axioms as Fields except that we do **not** require that every non-zero element has an inverse.

Definition 4.4: Homomorphism of Rings

Let R, S be rings. A **homomorphism of rings** is a map $\varphi : R \rightarrow S$ s.t.:

- $\forall r_1, r_2 \in R : \varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2)$.
- $\forall r_1, r_2 \in R : \varphi(r_1 \cdot r_2) = \varphi(r_1) \cdot \varphi(r_2)$
- $\varphi(1) = 1$.
- $\varphi(0) = 0$.

Corollary 4.5: Ψ for an Endomorphism is an isomorphism of Rings

Let V be a vector space over K . Then $\text{End}(V) := \text{Hom}(V, V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication (product) in $\text{End}(V)$ is given by composition, i.e.,

$$T_1 \cdot T_2 := T_1 \circ T_2 \quad \forall T_1, T_2 \in \text{Hom}(V, V).$$

The unity is id_V . Moreover, if V is finite dimensional and $\dim V = n$, and $\mathcal{B} = (v_1, \dots, v_n)$ is a basis for V , then $\Psi_{\mathcal{B}}^{\mathcal{B}} : \text{Hom}(V, V) \rightarrow M_{n \times n}(K)$ is an isomorphism of rings

4.1 Dual Space

An important special case of $\text{Hom}(V, W)$ is when $W = K$.

Definition 4.6: Dual Space

Let V be a vector space over K . The **dual space** of V , is defined to be the vector space $V^* := \text{Hom}(V, K)$. The elements of V^* are linear maps $\ell : V \rightarrow K$, and are sometimes called **functionals** (on V).

Corollary 4.7: Dimension of the Dual Space

Let V be a finite dimensional vector space over K . Then $\dim V^* = \dim V$.

Let $V = K_{\text{col}}^n$. Then we can identify V^* with K_{col}^n , as follows: If $\ell \in K_{\text{row}}^n$, then ℓ defines a functional $f_{\ell} : K_{\text{col}}^n \rightarrow K$ by $f_{\ell}(v) := \ell \cdot v \in K$. The map $D : K_{\text{row}}^n \rightarrow (K_{\text{col}}^n)^*$ defined by $D(\ell) := f_{\ell}$ is a linear map and is an isomorphism.

Let V be a finite dimensional vector space over K and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V . $\forall 1 \leq i \leq n$,

define a functional $v_i^* \in V^*$, i.e., $v_i^* : V \rightarrow K$ linear map, as follows:

$$v_i^*(v_j) := \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Since $v_1, \dots, v_n$ form a basis for V , our definition determines uniquely a well-defined functional $v_i^* : V \rightarrow K$.

Lemma 4.8: The Dual Basis

The elements $v_1^, \dots, v_n^* \in V^*$ form a basis for V^* . This basis is called the **dual basis** to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$.*

Moreover,

$$\forall \ell \in V^* : \ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*.$$

Proof. Let $\ell \in V^*$. We claim that

$$\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*. \quad (4.1)$$

Or in other words, we claim that

$$\forall v \in V : \ell(v) = \sum_{i=1}^n \ell(v_i) \cdot v_i^*(v). \quad (4.2)$$

Indeed, both sides of Equation (4.1) are linear maps $V \rightarrow K$, so it's enough to check that Equation (4.2) holds for $v = v_1, \dots, v = v_n$, b.c. $v_1, \dots, v_n$ form a basis for V . And indeed, for $v = v_k$ we have

$$\sum_{i=1}^n \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$

The claim just proven shows that $v_1^*, \dots, v_n^*$ span V^* . Since $\dim V^* = \dim V = n$, it follows that $v_1^*, \dots, v_n^*$ form a basis for V^* . $\square$

Question: Let V be a finite dimensional vector space over K . Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V . Therefore, $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^* . What is the relation between $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[id_V]_{\mathcal{B}}^{\mathcal{C}}$.

Proposition 4.9: Relation between $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[id_V]_{\mathcal{B}}^{\mathcal{C}}$

$$[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([id_V]_{\mathcal{C}}^{\mathcal{B}})^T = \left(([id_V]_{\mathcal{B}}^{\mathcal{C}})^T \right)^{-1}.$$

4.2 The Dual Map

Definition 4.10: The Dual Map

Let V, W be vector spaces over K . Let $T : V \rightarrow W$ be a linear map. Define a new map

$$T^* : W^* \rightarrow V^*, \quad T^*(\ell) = \ell \circ T \quad \forall \ell \in W^*.$$

Then T^* is a linear map. It is called **the dual map** to T .

Proof. Let $\ell \in W^*$, i.e., $\ell : W \rightarrow K$ a linear map. Clearly

$$\ell \circ T : V \rightarrow K$$

is a linear map, b.c. both ℓ and T are linear. So, $T^*(\ell) := \ell \circ T \in V^*$. This shows that $T^* : W^* \rightarrow V^*$ indeed has as it's target V^* as claimed, see also the Image 4.2. We'll show now that T^* is a linear

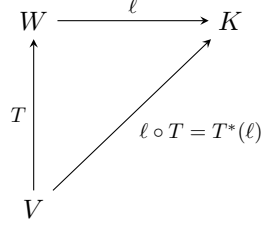


Figure 1: Map diagram of the dual map

map. Let $\ell \in W^*$, $\alpha \in K$. We have that $T^*(\alpha\ell) \in V^*$ is the map

$$\begin{aligned} T^*(\alpha\ell)(v) &= ((\alpha\ell) \circ T)(v) = (\alpha\ell)(Tv) = \alpha\ell(Tv) = \alpha(\ell \circ T)(v) = \alpha(T^*(\ell))(v) \\ \implies T^*(\alpha\ell) &= \alpha T^*(\ell). \end{aligned}$$

Now, let $\ell_1, \ell_2 \in W^*$. We have that $T^*(\ell_1 + \ell_2) \in V^*$ is the map

$$\begin{aligned} T^*(\ell_1 + \ell_2)(v) &= ((\ell_1 + \ell_2) \circ T)(v) = (\ell_1 \circ T + \ell_2 \circ T)(v) = (\ell_1 \circ T)(v) + (\ell_2 \circ T)(v) \\ &= (T^*(\ell_1))(v) + (T^*(\ell_2))(v) \\ \implies T^*(\ell_1 + \ell_2) &= T^*(\ell_1) + T^*(\ell_2). \end{aligned}$$

This shows that T^* is linear as claimed. □

Proposition 4.11: Composition of the Dual Map

Let U, V, W be vector spaces over K . Let $T : V \rightarrow W$, $S : W \rightarrow U$ be linear maps. Consider $T^* : W^* \rightarrow V^*$, $S^* : U^* \rightarrow W^*$, the dual maps. Then

$$(S \circ T)^* = T^* \circ S^*.$$

Lemma 4.12: Interpretation of the Matrix Transpose

Let $S : V \rightarrow W$ be a linear map between two finite dimensional vector spaces V and W over K . Let $\mathcal{B}$ be a basis for V and $\mathcal{C}$ be a basis for W . Consider $S^* : W^* \rightarrow V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^* respectively. Then

$$[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([S]_{\mathcal{C}}^{\mathcal{B}})^T.$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n), \mathcal{C} = (w_1, \dots, w_m)$. Define $A := [S]_{\mathcal{C}}^{\mathcal{B}}, A = (a_{ij})$. A satisfies

$$Sv_j = \sum_{i=1}^m a_{ij}w_i.$$

Now, for $w_j^* \in \mathcal{C}^*$ we have by Lemma 4.8

$$\begin{aligned} S^*w_j^* &= w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^* \\ &= \sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*. \end{aligned}$$

Therefore, the coordinates of $S^*w_j^*$ in the basis $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ are

$$\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix},$$

which is the row $\#j$ in A , represented as a column, or equivalently the column $\#j$ in the matrix A^T . $\square$

4.3 Reflexivity

Let V be a vector space over K . We have V^* , but also $V^{**} := (V^*)^*$ by applying 'dual' one more time, to V^* . $(V^*)^*$ is sometimes called the bidual space of V . Is there any relation between $(V^*)^*$ and V .

Theorem 4.13: The Canonical Isomorphism between $(V^*)^*$ and V

Let v be a *finite* dimensional vector space over K . Then there exists a canonical isomorphism $\tau : V \rightarrow (V^*)^*$, given by

$$\forall v \in V, \tau(v) : V^* \rightarrow K$$

is the map

$$\tau(v)(\ell) := \ell(v) \quad \forall \ell \in V^*.$$

In other words,

$$V \ni v \xrightarrow{\tau} (V^* \ni \ell \mapsto \ell(v) \in K).$$

Remark 4.14. The map $\tau : V \rightarrow (V^*)^*$ exists and is defined in the same way as above regardless of V being finite or infinite dimensional. Moreover, τ is always injective. However, when V is infinite dimensional, τ fails to be surjective.

4.4 Naturality

The map $\tau : V \rightarrow (V^*)^*$ exists for every space V , and is defined in the same way. To distinguish τ 's for different spaces, denote it now by $\tau_V : V \rightarrow (V^*)^*$. Let V, W be two vector spaces over K . Then for every linear map $T : V \rightarrow W$ we have

$$\begin{array}{ccc}
V & \xrightarrow{\tau_V} & (V^*)^* \\
\downarrow T & \text{\textcircled{C}} & \downarrow (T^*)^* \\
W & \xrightarrow{\tau_W} & (W^*)^*
\end{array}$$

which we call naturality.

Proposition 4.15: Image of τ

Let V be a finite dimensional vector space over K , let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V . Let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis and $(\mathcal{B}^*)^* = ((v_1^*)^*, \dots, (v_n^*)^*)$ the basis for $(V^*)^*$, which is dual to $\mathcal{B}^*$. Then $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, i.e.,

$$\tau(v_i) = (v_i^*)^* \quad \forall 1 \leq i \leq n.$$

4.5 Direct Sums

Proposition 4.16: The Vector Space of the Cartesian Product

Let V, W be a vector space over K . Then the cartesian product $V \times W$ becomes a vector space over K if we endow it with the following operations:

- $\forall (v_1, w_1), (v_2, w_2) \in V \times W : (v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2).$
- $\forall \alpha \in K, (v, w) \in V \times W : \alpha \cdot (v, w) := (\alpha v, \alpha w) \in V \times W.$
- $0_{V \times W} := (0_V, 0_W).$

This new vector space $V \times W$ is usually denoted $V \oplus W$ and is called the direct sum of V and W . We write the elements of $V \oplus W$ as (v, w) or $v \oplus w$.

Remark 4.17. $V \oplus W$ comes together with the following **canonical maps**:

- 1) $i_V : V \rightarrow V \oplus W, \quad i_V(v) := (v, 0).$
- 2) $i_W : W \rightarrow V \oplus W, \quad i_W(w) := (0, w).$
- 3) $p_V : V \oplus W \rightarrow V, \quad p_V(v, w) := v.$
- 4) $p_W : V \oplus W \rightarrow W, \quad p_W(v, w) := w.$

The blue marked maps are canonical embeddings, while the orange marked maps are canonical projections. These maps are all linear. The maps p_V, p_W are surjective, and i_V, i_W are injective. Because of the maps i_V, i_W we can sometimes view V and W as subspaces of $V \oplus W$.

Proposition 4.18: The map φ

Let V be a vector space over K , $U \subseteq V$ a subspace and $U' \subseteq V$ another subspace which is a complement of U in V (recall Definition 1.42). Then the map

$$\varphi : U \oplus U' \rightarrow V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

Remark 4.19. When U' is a complement of U in V , then $U + U' = V$ and we also have a canonical isomorphism $U \oplus U' \xrightarrow{\cong} V$. But formally speaking $U \oplus U'$ is a different vector space than V .

4.5.1 Generalization to more Summands

Let $U_1, \dots, U_m$ be **finitely** many vector spaces over K . We can define $U_1 \oplus \dots \oplus U_m := U_1 \times \dots \times U_m$, with the coordinatewise/componentwise operations.

For **infinite** families $U_i, i \in \mathcal{I}$ one can define

$$\bigoplus_{i \in \mathcal{I}} U_i := \left\{ t : \mathcal{I} \rightarrow \bigcup_{i \in \mathcal{I}} U_i \mid t(i) \in U_i \quad \forall i \in \mathcal{I}, t(i) \neq 0 \text{ for at most finitely many } i\text{'s} \right\}.$$

4.6 Quotient Spaces

Let V be a vector space over K , and $U \subseteq V$ a subspace. We will define a new vector space over K , V/U , together with a linear map $\pi : V \rightarrow V/U$ s.t. π is surjective and $\ker(\pi) = U$. V/U will have some other good properties. V/U is called the **quotient of V by U** or V **modulo U** . π is called the **quotient map** or the projection onto the quotient space.

We begin with an equivalence relation on v . Define for $v_1, v_2 \in V$

$$v_1 \sim v_2 \quad :\Longleftrightarrow \quad v_1 - v_2 \in U.$$

We claim that $\sim$ defines an equivalence relation on V . Indeed

- **Reflexivity:** $v \sim v$, b.c. $v - v = 0 \in U$.
- **Symmetry:** If $v_1 \sim v_2 \implies v_2 \sim v_1$, b.c. $v_2 - v_1 = -(v_1 - v_2) \in U$.
- **Transitivity:** If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 \sim v_3$, b.c. $v_1 - v_3 = \underbrace{(v_1 - v_2)}_{\in U} + \underbrace{(v_2 - v_3)}_{\in U} \in U$.

We denote the equivalence class of $v \in V$ by $[v]$. So

$$[v] = v + U = \{v + u \mid u \in U\},$$

or equivalently

$$[v] = \{w \in V \mid v \sim w\} = \{w \in V \mid v - w \in U\}.$$

Definition 4.20: Quotient Space

Let V be a vector space over K and $U \subseteq V$ a subspace. Define $V/U := V / \sim$ = the set of equivalence classes of the equivalence relation $\sim$. So, the elements of V/U are $[v]$, where $v \in V$. Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$. V/U is called the **quotient of V by U** (or modulo U).

We want to turn V/U into a vector space over K . To achieve this, we define:

- **Addition:** $\forall [v_1], [v_2] \in V/U, \quad [v_1] + [v_2] := [v_1 + v_2] \in V/U.$
- **Mult. by a scalar:** $\forall \alpha \in K, [v] \in V/U, \quad \alpha \cdot [v] := [\alpha v] \in V/U.$ (Alternatively written: $\alpha(v + U) := \alpha v + U.$)
- **The zero:** $0_{V/U} := [0_V].$ (Alternatively written: $0_{V/U} := 0_V + U.$)

Proposition 4.21: Well Definedness of The Operations

The operations above are well defined. Moreover, they give V/U the structure of a vector space over K .

We now define the map

$$\pi : V \rightarrow V/U, \quad v \mapsto [v] \in V/U.$$

Proposition 4.22: Properties of π

π is a linear map. Moreover, $\text{Im}(\pi) = V/U$ (i.e., π is surjective), and $\ker(\pi) = U$.

Proposition 4.23: Dimension of V/U

Suppose that V is finite dimensional. Then V/U is finite dimensional too, and $\dim V/U = \dim V - \dim U$.

Theorem 4.24: The Isomorphism Theorem

Let V, W be vector spaces over K and $T : V \rightarrow W$ a linear map. Define a new map $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$, as follows. Let $x \in V/\ker(T)$. Choose a representative $v \in V$ for x , i.e., $x = [v]$. Define $\bar{T}(x) = \bar{T}([v]) := T(v)$. Then:

1. $\bar{T}$ is well defined, and is a linear map.
2. $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$ is an isomorphism.
3. The Diagram in Figure 4.6 commutes, where $i : \text{Im}(T) \rightarrow W$ is the inclusion.

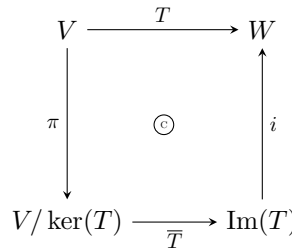


Figure 2: Map diagram of the map $\bar{T}$.

Proof. 1) $\bar{T}$ is well defined: Suppose $x = [v] = [v'] \in V/\ker(T)$, we need to show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$. Therefore, $T(v - v') = 0$. Since T is linear we conclude that $T(v) = T(v')$.

$\bar{T}$ is linear: Let $\alpha \in K, [v] \in V/\ker(T)$. By the linearity of T , the definition of the operations on the quotient space $V/\ker(T)$ and the definition of $\bar{T}$, we have that

$$\bar{T}(\alpha[v]) = \bar{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha \bar{T}([v]).$$

Now, let $[v_1], [v_2] \in V/\ker(T)$. Then

$$\bar{T}([v_1] + [v_2]) = \bar{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \bar{T}([v_1]) + \bar{T}([v_2]).$$

2) $\bar{T}$ is injective: Let $x \in \ker(\bar{T})$. Write x as $x = [v]$ for some $v \in V$. Then

$$\underbrace{\bar{T}(x)}_{=0} = T(v).$$

Therefore, $v \in \ker(T)$. Thus, $x = [v] = [0_V] = 0_{V/\ker(T)}$.

$\bar{T}$ is surjective: Let $w \in \text{Im}(T)$. Then there exists a $v \in V$ s.t. $T(v) = w$. Therefore, $\bar{T}([v]) = T(v) = w$. This shows that $\text{Im}(T) \subseteq \text{Im}(\bar{T})$.

But by the definition of $\bar{T}$ we also have that $\text{Im}(\bar{T}) \subseteq \text{Im}(T)$.

3) The diagram commutes: Let $v \in V$ and $w \in \text{Im}(T) \subseteq W$ s.t. $T(v) = w$. Then by the definition of $\pi, \bar{T}$ and i , we have that

$$(i \circ \bar{T} \circ \pi)(v) = (i \circ \bar{T})([v]) = i(T(v)) = i(w) = w.$$

□

4.6.1 Universal Property of Quotient Spaces

Theorem 4.25: Universal Property of Quotient Spaces

Let V be a vector space over K and $U \subseteq V$ a subspace. The quotient space V/U has the following universal property:

For every space W and every linear map $T : V \rightarrow W$, for which $T(U) = 0$ (i.e., for which $U \subseteq \ker(T)$), there exists a **unique** linear map $T' : V/U \rightarrow W$ s.t. the diagram in Figure 4.6.1 commutes. Moreover, $\ker(T') = \ker(T)/U$.

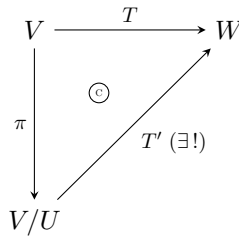


Figure 3: Map diagram of the map T' .

Jargon: Every linear map $T : V \rightarrow W$ that maps U to 0 factors through V/U (i.e., $T = T' \circ \pi$), and moreover in a unique way. We say that T **induces** a map $T' : V/U \rightarrow W$. We also say that T **descends** to a map $T' : V/U \rightarrow W$.

4.6.2 Quotient Spaces & Complements

Proposition 4.26: A Canonical Isomorphism to a Complement

Let V be a vector space over K and $U \subseteq V$ a subspace. Let $W \subseteq V$ be a complement of U . Then there exists a canonical isomorphism $Q : W \xrightarrow{\cong} V/U$, defined by

$$Q(w) := [w] \in V/U \quad \forall w \in W.$$

Remark 4.27. The isomorphism Q is canonical, only once W is specified. In general there is no canonical choice of a complement W to U . (In general there exist many choices of complements W to U , and there is no preferred choice.)

4.7 Relations to Dual Spaces

Proposition 4.28: Perpendicular Subspace

Let V be a vector space over K and $U \subseteq V$ a subspace. Define

$$U^\perp := \{\ell \in V^* \mid \ell|_U = 0\},$$

i.e., all the functionals on V that vanish on U . Then:

1. $U^\perp \subseteq V^*$ is a linear subspace.
2. There exists a **canonical** isomorphism $(V/U)^* \cong U^\perp$. If we assume that V is finite dimensional then there exists a **canonical** isomorphism $U^* \cong V^*/U^\perp$.

Proposition 4.29

Let $T : V \rightarrow W$ be a linear map. Then

$$(\text{Im}(T))^\perp = \ker(T^*).$$

5 Determinants

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(K).$$

There is a very simple criterion to check whether A is invertible or not.

Proposition 5.1: Criterion for Invertibility of a 2×2 Matrix

A is invertible if and only if $ad - bc \neq 0$. Moreover, if A is invertible, then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The number $ad - bc$ is $\det(A)$.

Definition 5.2: N-Linear Function

Let $D : M_{n \times n}(K) \rightarrow K$ be a function. We say that D is **n -linear** if for every $1 \leq i \leq n$, D is a linear function of the i -th row when the other $(n - 1)$ rows are held fixed.

We can think of $M_{n \times n}(K)$ as $K_{\text{row}}^n \times \dots \times K_{\text{row}}^n$ and if

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$$

we write $D(A) = D(\alpha_1, \dots, \alpha_n)$. Saying that D is n -linear means that for all $1 \leq i \leq n$, we have that

1.

$$\begin{aligned} D(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) &= D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n) \\ \forall \alpha_1, \dots, \alpha_i, \alpha'_i, \dots, \alpha_n &\in K_{\text{row}}^n \end{aligned}$$

2.

$$\begin{aligned} D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) &= c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) \\ \forall c \in K, \alpha_1, \dots, \alpha_i, \dots, \alpha_n &\in K_{\text{row}}^n \end{aligned}$$

Lemma 5.3: Linear Combinations of n -linear Functions

Any linear combination of n -linear functions is an n -linear function again.

Definition 5.4: Alternating Functions

Let $D : M_{n \times n}(K) \rightarrow K$ be an n -linear function. We say that D is **alternating** if for every matrix $A \in M_{n \times n}(K)$ in which some two rows are equal we have that $D(A) = 0$.

Proposition 5.5

Let D be an n -linear function. Assume that D has the property that whenever A has two equal adjacent (aufeinanderfolgende) rows then $D(A) = 0$. Then

1. *D is alternating.*
2. *For every matrix B , if B' is obtained from B by interchanging any two of its rows, then $D(B') = -D(B)$.*

Definition 5.6: Determinant Function

A function $D : M_{n \times n}(K) \rightarrow K$ is called **determinant function** if D is n -linear, is alternating and $D(I) = 1$.

Definition 5.7: Row&Column-Deletion Matrix

Let $A \in M_{n \times n}(K)$, $n \geq 2$, $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by

$$A(i|j) \in M_{(n-1) \times (n-1)}(K)$$

the matrix obtained from A by deleting row $\#i$ and column $\#j$.

If D is an $(n-1)$ -linear function and $A \in M_{n \times n}(K)$ we denote

$$D_{ij}(A) := D(A(i|j)).$$

Notation: Let $A \in M_{m \times n}(K)$ and $1 \leq i \leq m$, $1 \leq j \leq n$. Denote the element at position (i, j) in the matrix A by $A(i, j)$ or A_{ij} .

Theorem 5.8: Entwicklungssatz

Let $n \geq 2$ and D an $(n-1)$ -linear function, which is alternating. Let $1 \leq j \leq n$. Define a function $E_j : M_{n \times n}(K) \rightarrow K$ by

$$E_j(A) := \sum_{i=1}^n (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then E_j is an n -linear function and is alternating. If D is a determinant function then so is E_j .

Corollary 5.9: Existence of a Determinant Function

$\forall n \geq 1$, there exists at least one determinant function $M_{n \times n}(K) \rightarrow K$.

5.1 Uniqueness of the Determinant

Let $D : M_{n \times n}(K) \rightarrow K$ be an n -linear alternating function. Let

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix} \in M_{n \times n}(K).$$

Denote

$$I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix} = \begin{pmatrix} - & \varepsilon_1 & - \\ & \vdots & \\ - & \varepsilon_n & - \end{pmatrix}.$$

(So, $\varepsilon_1, \dots, \varepsilon_n$ is the standard basis of K_{row}^n .) Clearly,

$$\alpha_i = \sum_{j=1}^n A(i, j) \varepsilon_j \quad \forall 1 \leq i \leq n.$$

Therefore,

$$\begin{aligned} D(A) &= D(\alpha_1, \dots, \alpha_n) = D\left(\sum_{j=1}^n A(1, j) \varepsilon_j, \alpha_2, \dots, \alpha_n\right) = \sum_{j=1}^n A(1, j) \cdot D(\varepsilon_j, \alpha_2, \dots, \alpha_n) \\ &= \sum_{j=1}^n A(1, j) \cdot D\left(\varepsilon_j, \sum_{k=1}^n A(2, k) \varepsilon_k, \dots, \alpha_n\right) = \sum_{j,k=1}^n A(1, j) \cdot A(2, k) \cdot D(\varepsilon_j, \varepsilon_k, \dots, \alpha_n). \end{aligned}$$

Applying these steps to every row we get that

$$D(A) = \sum_{k_1, \dots, k_n=1}^n A(1, k_1) \cdot A(2, k_2) \cdot \dots \cdot A(n, k_n) \cdot D(\varepsilon_{k_1}, \varepsilon_{k_2}, \dots, \varepsilon_{k_n}), \quad (5.1)$$

where each of the indices $k_1, \dots, k_n$ runs between 1 and n . So far we haven't used the fact that D is alternating. Now, since D is alternating $D(\varepsilon_{k_1}, \dots, \varepsilon_{k_n}) = 0$, whenever two of the indices $k_1, \dots, k_n$ are equal. So, we are interested only in ordered sequences of indices $(k_1, \dots, k_n)$ in which $k_i \in \{1, \dots, n\} \quad \forall i$ and $k_i \neq k_j \quad \forall i \neq j$. Such a sequence is called a permutation of $\{1, \dots, n\}$, or a permutation of degree n .

Definition 5.10: Permutation

A **permutation** is a bijective map $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$.

So Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. Then the sequence $(k_1, \dots, k_n) = (\sigma 1, \dots, \sigma n)$ is a permutation as defined earlier. (Notation: $\sigma(i) := \sigma i$.) In other words, a permutation σ of degree n is a re-ordering of a list with n elements $(1, \dots, n) \xrightarrow{\sigma} (\sigma 1, \dots, \sigma n)$.

We can rewrite Equation (5.1) as

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \dots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}),$$

where S_n is the set of all permutations of degree n .

5.1.1 Important Thing About Permutations

Let σ be a permutation of degree n . Then one can pass from $(1, \dots, n)$ to $(\sigma 1, \dots, \sigma n)$ by performing a finite sequence of interchanges of pairs, called transpositions.

Definition 5.11: Transposition

A **transposition** is a permutation $\theta : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, where $\exists i, j \in \{1, \dots, n\}, i \neq j$ s.t. $\theta(i) = j, \theta(j) = i$ and $\theta(k) = k \quad \forall k \neq i, j$.

Now, every permutation $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ can be written as a composition of transpositions

$$\sigma = \theta_1 \circ \dots \circ \theta_m, \quad (5.2)$$

where $\theta_1, \dots, \theta_m$ are transpositions. But these θ_i 's and m are not unique. In other words, the same σ can be written as in Equation (5.2) in many different ways

$$\sigma = \theta_1 \circ \dots \circ \theta_m = \theta'_1 \circ \dots \circ \theta'_{m'}.$$

Why Can σ be written as in (5.2). Start with $(1, \dots, n)$ and if $\sigma 1 \neq 1$ then interchange 1 and

$\sigma 1$. We now have $(\sigma 1, \dots, 1, \dots, n)$. If $\sigma 2 \neq 2$, then interchange 1 and $\sigma 2$; otherwise proceed to 3, etc. But of course there are other ways to do it, e.g., $(3, 2, 1)$ can be obtained in 3 steps: $(1, 2, 3) \rightarrow (2, 1, 3) \rightarrow (2, 3, 1) \rightarrow (3, 2, 1)$. But also in one step $(1, 2, 3) \rightarrow (3, 2, 1)$.

Definition 5.12: Sign of a Permutation

Let σ be a permutation and let $\theta_1, \dots, \theta_m$ be transpositions s.t. $\sigma = \theta_1 \circ \dots \circ \theta_m$. The permutation σ is called **even** if m is even, and is called **odd** if m is odd. The number $(-1)^m \in \{-1, 1\}$ is called the **sign** of σ , i.e., $\text{sgn}(\sigma) := (-1)^m$.

Lemma 5.13: Well Definedness of the Sign of a Permutation

If $(\sigma 1, \dots, \sigma n)$ is obtained from $(1, \dots, n)$ in two different ways, one time by a sequence of m transpositions, and another time by a sequence of m' transpositions, then m' and m have the same parity, i.e., $(-1)^{m'} = (-1)^m$.

(So the parity of m depends only on σ and not on the particular way we performed a sequence of transpositions $(1, \dots, n) \rightarrow \dots \rightarrow (\sigma 1, \dots, \sigma n)$.)

Proof. We know that determinant functions $M_{n \times n}(K) \rightarrow K$ exists for every $n \geq 1$. Fix a determinant function $E : M_{n \times n}(K) \rightarrow K$. Consider $E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n})$. If $(\sigma 1, \dots, \sigma n)$ can be obtained from $(1, \dots, n)$ by a sequence of m transpositions, then the matrix

$$\begin{pmatrix} - & \varepsilon_1 & - \\ & \vdots & \\ - & \varepsilon_n & - \end{pmatrix}$$

can be obtained from

$$I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$$

by a sequence of m interchanges of pairs of rows. Therefore,

$$E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}) = (-1)^m E(\varepsilon_1, \dots, \varepsilon_n) = (-1)^m E(I) = (-1)^m.$$

But this applies also to m' . So,

$$E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}) = (-1)^{m'} E(\varepsilon_1, \dots, \varepsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

So we have that

$$(-1)^m = (-1)^{m'}.$$

□

5.1.2 Back to our $D(A)$

We have seen that if D is n -linear and alternating then

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \dots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}).$$

But

$$D(\varepsilon_{\sigma 1}, \varepsilon_{\sigma n}) = (\operatorname{sgn} \sigma) \cdot D(\varepsilon_1, \dots, \varepsilon_n) = (\operatorname{sgn} \sigma) \cdot D(I) \quad (5.3)$$

Theorem 5.14: Uniqueness of the Determinant Function

$\forall n \geq 1$, there exists a unique determinant function $M_{n \times n}(K) \rightarrow K$. We will denote it by $\det$. It is given by the formula

$$\det(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdots A(n, \sigma n).$$

Corollary 5.15

For every n -linear and alternating function $D : M_{n \times n}(K) \rightarrow K$ we have that

$$D(A) = \det(A) \cdot D(I) \quad \forall A \in M_{n \times n}(K).$$

5.1.3 A Bit more on Permutations

Denote by S_n the set of permutations $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. S_n has the structure of a group:

- We can compose permutations, i.e., if $\sigma, \tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, then $\sigma \cdot \tau := \sigma \circ \tau$, where $\sigma \circ \tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$.
- The identity $id : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ is the neutral element $1 \in S_n$.
- The inverse σ^{-1} of σ , is given by the inverse of the function σ (recall that σ is bijective, so σ^{-1} does exist).

Also, note that $|S_n| = n!$

Lemma 5.16: Sign of the Composition of Permutations

$\forall \sigma, \tau \in S_n$ we have

$$\operatorname{sgn}(\sigma \cdot \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

Theorem 5.17: The Determinant of the Product of Two Matrices

$\forall A, B \in M_{n \times n}(K)$ we have that

$$\det(A \cdot B) = \det(A) \cdot \det(B).$$

Proof. Fix a matrix $B \in M_{n \times n}(K)$, and consider a function $D : M_{n \times n}(K) \rightarrow K$ defined by

$$D(A) := \det(A \cdot B).$$

Claim: D is n -linear and alternating.

Proof of Claim: Write

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}.$$

If we view α_i as a $1 \times n$ matrix, then $\alpha_i B$ is also $1 \times n$, and in fact

$$A \cdot B = \begin{pmatrix} - & \alpha_1 B & - \\ & \vdots & \\ - & \alpha_n B & - \end{pmatrix}.$$

So,

$$D(A) = \det(\alpha_1 B, \dots, \alpha_n B).$$

Clearly, for every two row vectors α_i, α'_i we have that $(\alpha_i + \alpha'_i)B = \alpha_i B + \alpha'_i B$ and

$$\forall c \in K : (c \cdot \alpha_i)B = c \cdot (\alpha_i B).$$

Since $\det$ is n -linear it follows that D is also n -linear.

If $\alpha_i = \alpha_j$, then $\alpha_i B = \alpha_j B$. Therefore,

$$D(A) = \det(\alpha_1 B, \dots, \alpha_n B) = 0.$$

This proves the claim.

We continue with the proof of the Theorem. Since we found that D is n -linear and alternating, by Corollary 5.15, we have that

$$D(A) = \det(A) \cdot D(I).$$

But $D(I) = \det(I \cdot B) = \det(B)$. Therefore,

$$D(A) = \det(A \cdot B) = \det(A) \cdot \det(B). \quad \square$$

Corollary 5.18: Equivalent Condition for The Invertibility of a Matrix

If A is invertible then $\det(A) \neq 0$. Moreover,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

5.2 More on Determinants

Theorem 5.19: Determinant of a Transposed Matrix

$$\forall A \in M_{n \times n}(K) : \det(A^T) = \det(A).$$

Corollary 5.20

If $D : M_{n \times n}(K) \rightarrow K$ is an alternating n -linear function (of the rows of a matrix) then D is also an alternating and n -linear function of the columns of a matrix. In fact, $D(A^T) = D(A) \quad \forall A \in M_{n \times n}(K)$. In particular, this holds also for $D = \det$.

Theorem 5.21: Determinant after Row/Column Operations

1. If B is obtained from A by adding a multiple of one row of A to another row (or by doing the analogous operation on columns), then $\det(B) = \det(A)$.
2. If $A \xrightarrow{R_i \leftrightarrow R_j} B$, $i \neq j$, (or the analogous operation on columns) then $\det(B) = -\det(A)$.
3. If $A \xrightarrow{c \cdot R_i \rightarrow R_i} B$, then $\det(B) = c \cdot \det(A)$.

5.2.1 Determinants of Block Matrices

Consider a block matrix M of the type

$$M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where $A \in M_{r \times r}(K)$, $C \in M_{s \times s}(K)$, $B \in M_{r \times s}(K)$.

Proposition 5.22: Determinant of a Block Matrix

Let M be as above. Then

$$\det(M) = \det(A) \cdot \det(C).$$

Corollary 5.23: Entwicklungssatz with Cofactors

$\forall A \in M_{n \times n}(K)$ we have the following n different formulas for $\det(A)$, i.e.,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} \det(A(i|j)), \quad j = 1, \dots, n.$$

The scalars $c_{ij} := (-1)^{i+j} \det(A(i|j))$ are called the i, j **cofactor** of A . We have

$$\det(A) = \sum_{i=1}^n A_{ij} \cdot c_{ij}. \tag{5.4}$$

Claim: If we replace A_{ij} in Equation (5.4) by A_{ik} , where k is fixed, we always get 0. In other words, $\forall A \in M_{n \times n}(K)$, $\forall j$ we have

$$\sum_{i=1}^n A_{ik} \cdot c_{ij} = \begin{cases} \det(A) & j = k \\ 0 & j \neq k \end{cases}.$$

Proof of Claim: Let $B \in M_{n \times n}(K)$ be the matrix obtained from A by replacing column j of A by column k of A . So, in B column j equals column k . Clearly, $\det(B) = 0$ since B has two equal columns ($\iff B^T$ has two equal rows).

$$\implies 0 = \det(B) = \sum_{i=1}^n (-1)^{i+j} B_{ij} \det(B(i|j)) = \sum_{i=1}^n = \sum_{i=1}^n A_{ik} \det(A(i|j)) = \sum_{i=1}^n A_{ik} \cdot c_{ij}.$$

This proves the claim.

So, $\forall A \in M_{n \times n}(K)$, $\forall j, k$ we have

$$\sum_{i=1}^n A_{ik} \cdot c_{ij} = \delta_{jk} \cdot \det(A). \quad (5.5)$$

Definition 5.24: Classical Adjoint of A

The matrix $(c_{ij})^T$ is called the **classical adjoint of A** , i.e.,

$$\text{adj}(A) \in M_{n \times n}(K), \quad (\text{adj}(A))_{ij} = c_{ji} = (-1)^{i+j} \cdot \det(A(j|i)).$$

The Equation (5.5) says that

$$(\text{adj } A) \cdot A = \det(A) \cdot I.$$

Indeed,

$$\begin{aligned} \det(A) \cdot \delta_{jk} &= \sum_{i=1}^n c_{ij} \cdot A_{ik} = \sum_{i=1}^n (\text{adj}(A))_{ji} \cdot A_{ik} \\ \implies \det(A) \cdot I &= \text{adj}(A) \cdot A. \end{aligned}$$

Theorem 5.25: Formula for the inverse of A

Let $A \in M_{n \times n}(K)$. Then A is invertible if and only if $\det(A) \neq 0$. When A is invertible we have

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

5.3 Cramer's Rule

Let $A \in M_{n \times n}(K)$. We would like a formula to solve the system of equations $Ax = b$, where

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

are the unknowns and

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$$

is given. Assume $Ax = b$. Then

$$\begin{aligned} \text{adj } A \cdot Ax &= \text{adj } A \cdot b \\ \implies \det(A) \cdot x &= \text{adj } A \cdot b \\ \implies \forall 1 \leq j \leq n : \det(A) \cdot x_j &= \sum_{i=1}^n (\text{adj } A)_{ji} \cdot b_i = \sum_{i=1}^n (-1)^{i+j} b_i \det(A(i|j)), \end{aligned}$$

where the last expression is the determinant of the matrix obtained from A by replacing column j by b .

Theorem 5.26: Cramer's Rule

Let $A \in M_{n \times n}(K)$ with $\det(A) \neq 0$. Let $b \in K^n$. Then the system $Ax = b$ has a unique solution, and it is given by the formula

$$x_j = \frac{\det(A(j, b))}{\det(A)} \quad \forall 1 \leq j \leq n,$$

where $A(j, b)$ is the matrix obtained from A by replacing column j by b .

5.4 Determinants of Endomorphisms

Let $A \in M_{n \times n}(K)$ and $P \in GL_n(K)$, then

$$\det(P^{-1}AP) = \det(P^{-1}) \cdot \det(A) \cdot \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

In other words, similar matrices have the same determinant.

Let V be a finite dimensional vector space over K . Put $n := \dim V$. Let $T : V \rightarrow V$ be a linear map. Pick a basis $\mathcal{B}$ for V . We have a matrix $[T]_{\mathcal{B}}^{\mathcal{B}} \in M_{n \times n}(K)$. Consider $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$.

Question: Does $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ depend on the basis $\mathcal{B}$?

Answer: Let $\mathcal{B}'$ be another basis for V . Denote by $P := [id_V]_{\mathcal{B}}^{\mathcal{B}'}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'},$$

and we also have $[id_V]_{\mathcal{B}'}^{\mathcal{B}} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$. So,

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P. \quad \implies \quad \det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}.$$

Recall also that if $T, S : V \rightarrow V$ are two linear maps then $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$. Recall also that if $T : V \rightarrow V$ is an isomorphism then $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^{-1}$. Finally, $[id_V]_{\mathcal{B}}^{\mathcal{B}} = I$.

Corollary 5.27: Well Definedness of the Determinant of an Endomorphism

Let V be a finite dimensional vector space over K . Let $T : V \rightarrow V$ be a linear map. Then $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ does **not** depend on the choice of basis $\mathcal{B}$ of V . We denote this scalar by $\det(T) \in K$. The function

$$\det : \text{End}(V) \rightarrow K, \quad T \mapsto \det(T)$$

has the following properties:

1. $\det(S \circ T) = \det(S) \cdot \det(T) \quad \forall S, T \in \text{End}(V)$.
2. T is an isomorphism if and only if $\det(T) \neq 0$. If T is an isomorphism then

$$\det(T^{-1}) = \frac{1}{\det(T)}.$$

3. $\det(id_V) = 1$.
4. $\det(0) = 0 \in K$.

Remark 5.28. In contrast to $T : V \rightarrow V$, if we take $T : V \rightarrow W$, and $\mathcal{B}, \mathcal{C}$ bases of V, W , then $\det[T]_{\mathcal{C}}^{\mathcal{B}}$ **depends** on $\mathcal{B}, \mathcal{C}$ (unless $W = V$ and $\mathcal{C} = \mathcal{B}$).

6 Eigenvalues & Eigenvectors

Definition 6.1: Eigenvalues & Eigenvectors

Let V be a vector space over K and $T : V \rightarrow V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an **eigenvalue** of T if $\exists v \in V \setminus \{0\}$ s.t.

$$Tv = \lambda v.$$

Any vector $v \in V \setminus \{0\}$ with the property that $Tv = \lambda v$ is called an **eigenvector** of T for the eigenvalue λ .

Definition 6.2: Diagonalizability

Let V be a finite dimensional vector space over K and $T : V \rightarrow V$ a linear map. We say that T is diagonalizable if there exists a basis $\mathcal{B}$ of V s.t.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}. \quad (6.1)$$

Lemma 6.3: Equivalent Condition for Diagonalizability

T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall i, Tv_i = \lambda_i v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape as in Equation (6.1), then $\forall i, \lambda_i$ is an eigenvalue of T and v_i is an eigenvector of λ_i .

Remark 6.4. We saw that $\forall A \in M_{n \times n}(K)$ we have a linear map $T_A : K^n \rightarrow K^n$, $T_A(v) := Av$. We say that $A \in M_{n \times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.

Proposition 6.5: Linear Independence of Eigenvectors of Distinct Eigenvalues

Let V be a vector space over K and $T : V \rightarrow V$ a linear map. Suppose that $\lambda_1, \dots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j \ \forall i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent.

Proof. Induction on n .

$n = 1$: v_1 is an eigenvector for λ_1 . By definition of $v_1 \neq 0 \implies v_1$ is linearly independent.

Induction step: Let $n \geq 1$ and suppose the proposition holds for any list of n eigenvalues and eigenvectors. We'll prove now that the proposition holds also for a list of $n + 1$ eigenvalues and eigenvectors. Let $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T , and $v_1, \dots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$, respectively. We need to prove that $v_1, \dots, v_n, v_{n+1}$ are linearly independent.

Assume that

$$a_1 v_1 + \dots + a_n v_n + a_{n+1} v_{n+1} = 0. \quad (6.2)$$

We need to show that $a_1 = \dots = a_n = a_{n+1} = 0$. Apply T to Equation (6.2). Then we have that

$$a_1 T v_1 + \dots, a_n T v_n + a_{n+1} T v_{n+1} = 0.$$

But $\forall i : T v_i = \lambda_i v_i$. Therefore,

$$a_1 \lambda_1 v_1 + \dots, a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0. \quad (6.3)$$

Consider now Equation (6.3) - $\lambda_{n+1} \cdot (6.2)$, i.e.,

$$a_1 (\lambda_1 - \lambda_{n+1}) v_1 + \dots, a_n (\lambda_n - \lambda_{n+1}) v_n + a_{n+1} \underbrace{(\lambda_{n+1} - \lambda_{n+1})}_{=0} v_{n+1} = 0.$$

By the induction hypothesis $v_1, \dots, v_n$ are linearly independent. Therefore,

$$a_1 (\lambda_1 - \lambda_{n+1}) = \dots, a_n (\lambda_n - \lambda_{n+1}) = 0.$$

By assumption $\lambda_i \neq \lambda_{n+1} \quad \forall 1 \leq i \leq n$. So we have that $a_1 = \dots = a_n = 0$. Going back to Equation (6.2) we get that $a_{n+1} \cdot v_{n+1} = 0$. But by assumption we also have that $v_{n+1} \neq 0$. Therefore, $a_{n+1} = 0$. $\square$

6.1 The Characteristic Polynomial

Proposition 6.6: Equivalent Condition for Eigenvalue

*Let V be a vector space over K and $T : V \rightarrow V$ a linear map. Then $\lambda \in K$ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \text{id}$ is **not** injective.*

Definition 6.7: The Characteristic Polynomial

Let V be a finite dimensional vector space over K and $T : V \rightarrow V$ a linear map. We define the **characteristic polynomial of T** as

$$p_T(x) := \det(T - x \cdot \text{id}_V).$$

We define the characteristic polynomial of a matrix $A \in M_{n \times n}(K)$ in the same way, i.e.,

$$p_A(x) := \det(A - x \cdot I_n).$$

Definition 6.8: Eigenspace

Let V be a vector space over K and $T \in \text{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T . The **eigenspace** corresponding to λ is defined as

$$\text{Eig}_T(\lambda) := \{v \in V \mid T v = \lambda v\} = \ker(T - \lambda \text{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly, since $\text{Eig}_T(\lambda)$ is the kernel of a linear map, $\text{Eig}_T(\lambda) \subseteq V$ is a linear subspace of V .

Lemma 6.9: Properties of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$. Then:

1. $p_A(x)$ is a polynomial of degree n , with leading coefficient $(-1)^n$, i.e., $p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$, with $c_i \in K \quad \forall i$, and $c_n = (-1)^n$.
2. $p_{A^T}(x) = p_A(x)$.
3. If A has upper triangular form as in Definition 3.12, then

$$p_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x).$$

4. If

$$A = \begin{pmatrix} B_1 & * \\ 0 & B_2 \end{pmatrix}$$

is a block matrix, then

$$p_A(x) = p_{B_1}(x) \cdot p_{B_2}(x).$$

5. If A and B are similar matrices, i.e., $B = P^{-1}AP$ for some $P \in GL_n(K)$, then

$$p_A(x) = p_B(x).$$

Remark 6.10. If V is an n -dimensional vector space over K and $T \in \text{End}(V)$, then $p_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Definition 6.11: The Trace

Let $A \in M_{n \times n}(K)$. Define the **trace** (Spur) of A as

$$\text{Tr}(A) := \sum_{i=1}^n A_{ii} \in K,$$

i.e., the sum of the entries along the diagonal of A . If V is a finite dimensional vector space over K and $T \in \text{End}(V)$, we define the trace of T as

$$\text{Tr}(T) := \text{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}),$$

where $\mathcal{B}$ is some basis for V . We'll see soon that $\text{Tr}(T)$ is well defined, i.e., it does not depend on the choice of $\mathcal{B}$.

Lemma 6.12: Properties of the Trace

1.

$$\forall A, B \in M_{n \times n}(K) : \operatorname{Tr}(A \cdot B) = \operatorname{Tr}(B \cdot A).$$

2. If A and B are similar matrices, then

$$\operatorname{Tr}(A) = \operatorname{Tr}(B).$$

In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V , then

$$\operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \operatorname{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'}).$$

Thus, $\operatorname{Tr}(T)$ is well defined.

Proof. 1):

$$\begin{aligned} \operatorname{Tr}(A \cdot B) &= \sum_{i=1}^n (A \cdot B)_{ii} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} \cdot B_{ji} \\ &= \sum_{j=1}^n \sum_{i=1}^n A_{ij} \cdot B_{ji} = \sum_{j=1}^n \sum_{i=1}^n B_{ji} A_{ij} = \sum_{j=1}^n (B \cdot A)_{jj} \\ &= \operatorname{Tr}(B \cdot A). \end{aligned}$$

2): If $B = P^{-1}AP$ for some $P \in GL_n(K)$, then by 1) we have that

$$\operatorname{Tr}(B) = \operatorname{Tr}(P^{-1}AP) = \operatorname{Tr}(P^{-1} \cdot (AP)) = \operatorname{Tr}((AP) \cdot P^{-1}) = \operatorname{Tr}(A).$$

Regarding the claim about $\operatorname{Tr}(T)$, recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} \quad (6.4)$$

and $[id_V]_{\mathcal{B}'}^{\mathcal{B}} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$. So, by Equation (6.4) we have that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [T]_{\mathcal{B}}^{\mathcal{B}}. \quad \square$$

Lemma 6.13: Coefficients of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$ and $p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$ its characteristic polynomial. We've already seen that $c_n = (-1)^n$. We also have that

$$c_{n-1} = (-1)^{n-1} \cdot \operatorname{Tr}(A), \quad c_0 = \det(A).$$

In other words,

$$p_A(x) = (-1)^n x^n + (-1)^{n-1} x^{n-1} + c_{n-2} x^{n-2} + \dots + c_1 x + \det(A).$$

Definition 6.14: Direct Sum

Let V be a vector space over K and $U_1, \dots, U_r \subseteq V$ subspaces. Denote $W := U_1 + \dots + U_r \subseteq V$. We say that W is a **direct sum** of $U_1, \dots, U_r$ (or that $U_1, \dots, U_r$ is a direct sum) if every $w \in W$ has a **unique** representation as $w = u_1 + \dots + u_r$ with $u_i \in U_i \quad \forall i$.

Proposition 6.15: Eigenspaces are a Direct Sum

Let V be a vector space over K and $T \in \text{End}(V)$. Let $\lambda_1, \dots, \lambda_r \in K$ be pairwise distinct eigenvalues of T . Then

$$\text{Eig}_T(\lambda_1) + \dots + \text{Eig}_T(\lambda_r)$$

is a direct sum.

Corollary 6.16: Equivalent Condition for Diagonalizability of an Endomorphism

Assume V is finite dimensional, $T \in \text{End}(V)$ and let $\lambda_1, \dots, \lambda_r \in K$ be all the eigenvalues of T . Assume additionally that $\lambda_1, \dots, \lambda_r$ are pairwise distinct. Then T is diagonalizable if and only if

$$\sum_{i=1}^r \dim \text{Eig}_T(\lambda_i) = \dim V.$$

Lemma 6.17: Decomposition of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$ (or $T \in \text{End}(V)$) be diagonalizable. Then $p_A(x)$ decomposes as a product of linear factors, i.e., degree 1

$$p_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x),$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i 's).

6.1.1 A Quick Digression about Polynomials

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$ we can write

$$p(x) = (x - \lambda) \cdot q(x) + R,$$

where $\deg q(x) = \deg p(x) - 1$ and $R \in K$. But if we substitute $x = \lambda$, we obtain

$$p(\lambda) = 0 \cdot q(\lambda) + R = R \implies p(x) = (x - \lambda) \cdot q(x) + p(\lambda).$$

Conclusion: $p(\lambda) = 0 \iff p(x) = (x - \lambda) \cdot q(x)$ for some $q(x) \in K[x]$. We say that λ is a zero of $p(x)$ (Nullstelle) or a root of $p(x)$.

Sometimes λ is also a zero of $q(x)$, so $q(x) = (x - \lambda) \cdot h(x)$. Therefore, $p(x) = (x - \lambda)^2 \cdot h(x)$, etc.

Definition 6.18: Multiplicity of Zeroes

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. We say that λ is a **zero of multiplicity** $m \in \mathbb{N}_{>0}$ if

$$p(x) = (x - \lambda)^m \cdot g(x),$$

where $g(x) \in K[x]$ and $g(\lambda) \neq 0$.

If $m \geq 2$ we say λ is a **multiple zero** of $p(x)$.

If $m = 1$ we say λ is a **simple zero** of $p(x)$.

6.2 Geometric & Algebraic Multiplicities of Eigenvalues**Definition 6.19: Geometric & Algebraic Multiplicities**

Let V be a vector space over K and $T \in \text{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T . We define the **geometric multiplicity** of λ to be

$$m_g(T; \lambda) := \dim \text{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite dimensional. We define the **algebraic multiplicity** of λ to be the multiplicity of the zero λ in the characteristic polynomial $p_T(x)$ of T . We denote it by $m_A(T; \lambda)$.

Proposition 6.20: Inequality Regarding Geometric & Algebraic Multiplicity

$$m_g(T; \lambda) \leq m_A(T; \lambda).$$

Proof. Let $v_1, \dots, v_k$ be a basis for $\text{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V . Then we have

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} | & & | & & | & & | \\ [Tv_1]_{\mathcal{B}} & \dots & [Tv_n]_{\mathcal{B}} & [Tw_{k+1}]_{\mathcal{B}} & \dots & [Tw_n]_{\mathcal{B}} \\ | & & | & & | & & | \end{pmatrix} = \begin{pmatrix} \lambda & \dots & 0 & * \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & \lambda & * \\ 0 & \dots & 0 & C \end{pmatrix},$$

where $C \in M_{(n-k) \times (n-k)}(K)$. Therefore,

$$p_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - xI_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k}) = (\lambda - x)^k \cdot p_C(x) = (-1)^k \cdot (x - \lambda)^k \cdot p_C(x).$$

this shows that the multiplicity of λ as a zero of $p_T(x)$ is at least k . In other words,

$$m_A(T; \lambda) \geq k = m_g(T; \lambda).$$

□

6.3 Diagonalizability

Theorem 6.21: Equivalent Statements for Diagonalizability

Let V be an n -dimensional vector space over K and $T \in \text{End}(V)$. The following statements are equivalent:

1. T is diagonalizable.
2. There exists a basis for V whose elements are eigenvectors of T .
3. The characteristic polynomial $p_T(x)$ splits into a product of linear factors, and for every eigenvalue of T the geometric and algebraic multiplicities are equal.

4.

$$\sum_{i=1}^k \dim \text{Eig}_T(\lambda_i) = \dim V,$$

where $\lambda_1, \dots, \lambda_k \in K$ are the eigenvalues of T .

5.

$$V \cong \bigoplus_{i=1}^k \text{Eig}_T(\lambda_i).$$

Corollary 6.22

If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

Corollary 6.23

If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

6.4 Trigonalization

Definition 6.24: Trigonalizability

We say that $T \in \text{End}(V)$ is **trigonalizable** if there exists a basis $\mathcal{B}$ for V s.t. $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & * \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}.$$

Theorem 6.25: Equivalent Condition for Trigonalizability

Let V be a finite dimensional vector space over K and $T \in \text{End}(V)$. Then T is trigonalizable if and only if $p_T(x)$ splits as a product of linear factors, i.e.,

$$p_T(x) = (\lambda_1 - x)^{l_1} \cdots (\lambda_k - x)^{l_k}.$$

Theorem 6.26: Fundamental Theorem of Algebra

Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients, splits as a product of linear factors and a constant, i.e.,

$$p(x) = c \cdot (x - a_1)^{n_1} \cdot \dots \cdot (x - a_r)^{n_r},$$

with $c \in \mathbb{C}$, $r \geq 0$, $a_i \in \mathbb{C}$, $n_j \in \mathbb{N}_{>0}$.

Corollary 6.27: Trigonalizability over $\mathbb{C}$

Let V be a finite dimensional vector space over $\mathbb{C}$. Then every $T \in \text{End}(V)$ is trigonalizable.

6.5 Minimal Polynomial & Cayley-Hamilton Theorem**Definition 6.28: Matrix Polynomial**

Let $A \in M_{n \times n}(K)$. Let $g(x) = a_d x^d + \dots + a_1 x + a_0 \in K[x]$ be a polynomial. We can define $g(A) \in M_{n \times n}(K)$, as

$$g(A) := a_d A^d + \dots + a_1 A + a_0 I_n.$$

Remark 6.29. $\forall A \in M_{n \times n}(K) \exists g \in K[x]$ s.t. $g(A) = 0$. Indeed, consider

$$I_n, A, A^2, \dots, A^r \in M_{n \times n}(K). \quad (6.5)$$

Since $M_{n \times n}(K)$ is a vector space of dimension n^2 , then if we take $r > n^2$ then the elements in (6.5) cannot be linearly independent. Therefore, $\exists a_0, a_1, \dots, a_r \in K$ s.t.

$$a_0 I_n + a_1 A + \dots + a_r A^r = 0.$$

A similar thing applies to $T \in \text{End}(V)$, where V is a finite dimensional vector space over K .
 $(T^k = \underbrace{T \circ \dots \circ T}_{\times k}, g(T) := a_d T^d + \dots + a_1 T + a_0 \text{id}_V.)$

Definition 6.30: Minimal Polynomial

Let V be a vector space over K and $T \in \text{End}(V)$. A **minimal polynomial** for T is a polynomial $g(x) \in K[x]$ of the minimal possible degree s.t. $g(T) = 0$.

Theorem 6.31: Cayley-Hamilton

1. Let $A \in M_{n \times n}(K)$. Then

$$p_A(A) = 0.$$

2. Let V be a finite dimensional vector space over K , and $T \in \text{End}(V)$. Then

$$p_T(T) = 0.$$

The proof of this Theorem may be important for the exam but will be skipped in this Document. The proof can be found in the lecture notes.

7 Euclidean & Hermitian (unitary) Spaces

At last we endow our spaces with a geometric structure. We work here with vector spaces V over $K = \mathbb{R}$ or $K = \mathbb{C}$. Over $\mathbb{R}$ we'll have Euclidean spaces. Over $\mathbb{C}$ we'll have Hermitian (a.k.a unitary) spaces.

Definition 7.1: Scalar Product

Let V be a vector space over $\mathbb{R}$. A **scalar product** (a.k.a inner product) is a function $V \times V \rightarrow \mathbb{R}$, $(v, w) \mapsto \langle v, w \rangle$ that has the following properties:

1. Linearity in the first variable:

$$\begin{aligned}\langle v_1 + v_2, w \rangle &= \langle v_1, w \rangle + \langle v_2, w \rangle & \forall v_1, v_2, w \in V \\ \langle \alpha v, w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{R}, v, w \in V.\end{aligned}$$

2. Linearity in the second variable:

$$\begin{aligned}\langle v, w_1 + w_2 \rangle &= \langle v, w_1 \rangle + \langle v, w_2 \rangle & \forall v, w_1, w_2 \in V \\ \langle v, \alpha w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{R}, v, w \in V.\end{aligned}$$

3. Symmetry:

$$\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V.$$

4. Positivity:

$$\langle v, v \rangle > 0 \quad \forall v \in V \setminus \{0\}.$$

We call $(V, \langle \cdot, \cdot \rangle)$ an **Euclidean space**, or a space endowed with a scalar product.

Definition 7.2: Hermitian Product

Let V be a vector space over $\mathbb{C}$. A **Hermitian product** (or inner product) on V is a function $V \times V \rightarrow \mathbb{C}$, $(v, w) \mapsto \langle v, w \rangle$ with the following properties:

1. Linearity in the first variable:

$$\begin{aligned}\langle v_1 + v_2, w \rangle &= \langle v_1, w \rangle + \langle v_2, w \rangle & \forall v_1, v_2, w \in V \\ \langle \alpha v, w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{C}, v, w \in V.\end{aligned}$$

2. Sesquilinearity (complex anti-linearity) in the second variable:

$$\begin{aligned}\langle v, w_1 + w_2 \rangle &= \langle v, w_1 \rangle + \langle v, w_2 \rangle & \forall v, w_1, w_2 \in V \\ \langle v, \alpha w \rangle &= \bar{\alpha} \langle v, w \rangle & \forall \alpha \in \mathbb{C}, v, w \in V.\end{aligned}$$

3. Hermitian property:

$$\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \forall v, w \in V.$$

4. Positivity:

$$\langle v, v \rangle > 0 \quad \forall v \in V.$$

We call $(V, \langle \cdot, \cdot \rangle)$ a **Hermitian (or unitary) space**.

Lemma 7.3: Properties of Euclidean/Hermitian Spaces

Let V be an Euclidean/Hermitian vector space. Then:

1. $\forall v \in V : \langle 0, v \rangle = \langle v, 0 \rangle = 0$.
2. Let $w \in V$. If $\langle v, w \rangle = 0 \quad \forall v \in V$, then $w = 0$.
3. Let $w_1, w_2 \in V$. If $\langle v, w_1 \rangle = \langle v, w_2 \rangle \quad \forall v \in V$, then $w_1 = w_2$.

Definition 7.4: Some Definitions to Euclidean/Hermitian Spaces

1. Let V be an Euclidean/Hermitian vector space. $\forall v \in V$, define $\|v\| := \sqrt{\langle v, v \rangle} \in \mathbb{R}_{\geq 0}$. $\|\cdot\|$ is called the **norm** on V induced by $\langle \cdot, \cdot \rangle$.

2. A vector $u \in V$ is called a **unit vector** if $\|u\| = 1$. Note that if $0 \neq v \in V$ is any vector then

$$\frac{v}{\|v\|}$$

is a unit vector (which is positively proportional to v , so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the normalization of v .

3. For $v, w \in V$ we define the **distance** between v and w by

$$d(v, w) := \|v - w\|.$$

4. Two vectors $u, v \in V$ are called **orthogonal** (or perpendicular one to the other) if $\langle u, v \rangle = 0$. In this case we sometimes write $u \perp v$.

5. A subset $S \subseteq V$ is called an **orthogonal system** if $0 \notin S$ and for every two distinct elements $u, v \in S$ we have $u \perp v$.

6. An orthogonal system $S \subseteq V$ is called **orthonormal** if $\forall v \in S$ we have $\|v\| = 1$.

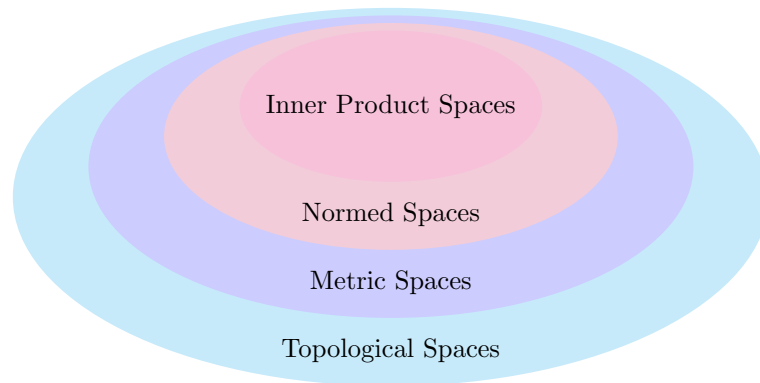


Figure 4: Overview of the generality of different spaces

Theorem 7.5: Pythagoras Theorem

Let V be an Euclidean/Hermitian vector space and $u, v \in V$ with $u \perp v$. Then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Lemma 7.6: Induced Norm

Let V be an Euclidean space. Then

$$\forall u, v \in V : \langle u, v \rangle = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

This means that $\|\cdot\|$ determines $\langle \cdot, \cdot \rangle$.

If V is an Hermitian space, we have that

$$\forall u, v \in V : \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

Warning: There exist norms that do **not** come from scalar products.

Definition 7.7: Complex Conjugated Matrix/Adjoint Matrix/Hermitian Matrix

1. Let $A \in M_{n \times m}(\mathbb{C})$. We write $\overline{A}$ for the matrix whose entry at i, j is $\overline{(A_{ij})}$, i.e.,

$$(\overline{A})_{ij} := \overline{(A_{ij})}.$$

2. Let $A \in M_{n \times n}(\mathbb{C})$. Define

$$A^* := \overline{A}^T.$$

A^* is called the **adjoint matrix** of A . (Has nothing to do with the classical adjoint $\text{adj } A$).

3. $A \in M_{n \times n}(\mathbb{C})$ is called **Hermitian** if $A^* = A$ ($\iff A^T = \overline{A}$).

Important Example: Let $A \in M_{n \times n}(\mathbb{R})$. Define

$$\langle \cdot, \cdot \rangle_A : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}, \quad \langle v, w \rangle_A = v^T A w.$$

We can quickly see that $\langle \cdot, \cdot \rangle_A$ is bilinear. Assume now that A is symmetric. Then $\langle v, w \rangle_A = \langle w, v \rangle_A \quad \forall v, w \in V$. Indeed,

$$\langle w, v \rangle_A = w^T A v = w^T A^T v = (v^T A w)^T = (\langle v, w \rangle_A)^T = \langle v, w \rangle_A,$$

where we've used that a scalar can be viewed as a 1×1 matrix.

Is $\langle \cdot, \cdot \rangle_A$ in general a scalar product?

In general it is not, e.g., for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we have that

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle_A = -1.$$

But for a scalar product we need that $\langle v, v \rangle > 0 \quad \forall v \neq 0$.

Definition 7.8: Positive Definite

A symmetric matrix $A \in M_{n \times n}(\mathbb{R})$ is called **positive definite** if

$$v^T A v > 0 \quad \forall v \in \mathbb{R}^n \setminus \{0\}.$$

Proposition 7.9: Scalar Product with Matrices

Let $A \in M_{n \times n}(\mathbb{R})$. Then $\langle \cdot, \cdot \rangle_A$ is a scalar product if and only if A is positive definite.

The situation for Hermitian products is similar.

Definition 7.10: Positive Definite for Hermitian Matrices

Let $A \in M_{n \times n}(\mathbb{C})$ be a Hermitian matrix. We say that A is **positive definite** if

$$v^T A \bar{v} > 0 \quad \forall v \in \mathbb{C}^n \setminus \{0\}.$$

Note that $v^T A \bar{v} \in \mathbb{R}$, b.c.

$$\overline{(v^T A \bar{v})} = \bar{v}^T \bar{A} v = (\bar{v}^T \bar{A} v)^T = v^T \bar{A}^T \bar{v} = v^T A \bar{v}.$$

Given a positive definite complex matrix A we can define a Hermitian product on $\mathbb{C}^n$, i.e.,

$$\langle v, w \rangle_A = v^T A \bar{w}.$$

Proposition 7.11: Hermitian Product with Matrices

Let $A \in M_{n \times n}(\mathbb{C})$. Then $\langle \cdot, \cdot \rangle_A$ is a Hermitian product if and only if A is positive definite.

7.1 Norms & Angles**Definition 7.12: Norm**

Let V be a vector space over K , where $K = \mathbb{R}$ or $\mathbb{C}$. A **norm** on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ s.t.:

1. Triangle inequality:

$$\|u + v\| \leq \|u\| + \|v\| \quad \forall u, v \in V.$$

2. Homogeneity:

$$\|\alpha v\| = |\alpha| \cdot \|v\| \quad \forall \alpha \in K, v \in V.$$

3. Non-degeneracy:

$$\|v\| = 0 \implies v = 0.$$

Theorem 7.13: Cauchy-Schwarz Inequality

Let $(V, \langle \cdot, \cdot \rangle)$ be a vector space endowed with an inner product $\langle \cdot, \cdot \rangle$. Then

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\| \quad \forall u, v \in V,$$

where equality holds if and only if u and v are linearly dependent.

Corollary 7.14: Induced Norm

Let V be a vector space and $\langle \cdot, \cdot \rangle$ an inner product on V , then

$$V \ni v \mapsto \|v\| := \sqrt{\langle v, v \rangle}$$

is a norm on V .

Definition 7.15: Angle

Let $(V, \langle \cdot, \cdot \rangle)$ be an Euclidean space. Let $u, v \in V$ be two non-zero vectors. By the Cauchy-Schwarz inequality 7.13, we have that

$$-1 \leq \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \leq 1.$$

We define the **angle** between u and v to be the unique $\alpha \in [0, \pi]$ s.t.

$$\cos \alpha = \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \left(= \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle \right)$$

7.2 The Gram-Schmidt Orthogonalization Process**Theorem 7.16: Properties of Orthogonal Systems**

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space.

1. If $S \subseteq V$ is an orthogonal system then S is linearly independent.
2. If $v_1, \dots, v_n$ is an orthogonal system and $v \in \text{Sp}(v_1, \dots, v_n)$ and write $v = a_1 v_1 + \dots + a_n v_n$, for $a_j \in K$. Then the coefficients a_j are given by

$$a_j = \frac{\langle v, v_j \rangle}{\langle v_j, v_j \rangle} \quad \forall 1 \leq j \leq n.$$

Corollary 7.17: Matrix for an Orthonormal Basis

Let V, W be a vector space over $K = \mathbb{R}$ or $\mathbb{C}$ and let $\langle \cdot, \cdot \rangle$ be an inner product on W . Let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V and $\mathcal{C} = (w_1, \dots, w_m)$ be an orthonormal basis for W . Let $T : V \rightarrow W$ be a linear map. Write $[T]_{\mathcal{C}}^{\mathcal{B}} = (a_{ij})$. Then

$$a_{ij} = \langle T v_j, w_i \rangle \quad \forall i, j.$$

(If $\mathcal{C}$ is only a orthogonal basis then $a_{ij} = \langle T v_j, w_i \rangle / \langle w_i, w_i \rangle$).

So having an ortho-gonal/normal basis for our spaces is useful!

Theorem 7.18: Gram-Schmidt Orthogonalization Process

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$ and let $n = \dim V$. Let $(v_1, \dots, v_n)$ be a basis for V . Define a new list of vectors $w_1, \dots, w_n$ by induction, as follows

$$w_1 := v_1,$$

$$w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \cdot w_i \quad \text{for } j = 2, \dots, n.$$

Then:

1. $(w_1, \dots, w_n)$ is an orthogonal basis for V .
2. $(\frac{w_1}{\|w_1\|}, \dots, \frac{w_n}{\|w_n\|})$ is an orthonormal basis for V .
3. $\forall 1 \leq j \leq n$ we have $\text{Sp}(v_1, \dots, v_j) = \text{Sp}(w_1, \dots, w_j) = \text{Sp}(\frac{w_1}{\|w_1\|}, \dots, \frac{w_j}{\|w_j\|})$.

Corollary 7.19: Existence of an Orthonormal Basis for Finite Dimensional Spaces

Every finite dimensional inner product space has an orthonormal basis.

7.3 The Orthogonal Complement

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$.

Definition 7.20: Orthogonal Complement

Let $\emptyset \neq S \subseteq V$ be a subset. We define the **orthogonal complement of S** as

$$S^\perp := \{v \in V \mid v \perp s \quad \forall s \in S\} = \{v \in V \mid \langle v, s \rangle = 0 \quad \forall s \in S\}.$$

When $S = \{v\}$ we write $v^\perp := \{v\}^\perp$.

Lemma 7.21: The Orthogonal Complement is a Subspace

Let $\emptyset \neq S \subseteq V$ be a subset. Then $S^\perp$ is a linear subspace.

Proof. $0 \in S^\perp$ b.c. $\langle 0, s \rangle = 0$ for every $s \in S$. Furthermore, if $\alpha, \beta \in K$ and $v, w \in S^\perp$ then for every $s \in S$ we have that

$$\langle \alpha v + \beta w, s \rangle = \underbrace{\alpha \langle v, s \rangle}_{=0} + \underbrace{\beta \langle w, s \rangle}_{=0} = 0.$$

Therefore, $\alpha v + \beta w \in S^\perp$. □

Lemma 7.22: Further Properties of the Orthogonal Complement

Let $\emptyset \neq S, T \subseteq V$. Then:

1. $0^\perp = V$, $V^\perp = \{0\}$.
2. $S \cap S^\perp$ is either $\emptyset$ or equal to $\{0\}$.
3. If $S \subseteq T$ then $T^\perp \subseteq S^\perp$.
4. $(\text{Sp}(S))^\perp = S^\perp$.
5. $S \subseteq (S^\perp)^\perp$.

Theorem 7.23: The Orthogonal Complement is a Complement

Let V be an inner product space (possibly of infinite dimension), and $U \subseteq V$ a **finite dimensional** subspace. Then $U^\perp$ is a complement of U in V (, i.e., $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$).

Proof. By Lemma 7.22 2., we have that $U \cap U^\perp = \{0\}$, since $U \cap U^\perp \neq \emptyset$. So it's enough to show that $U + U^\perp = V$.

Let $r := \dim U$ and pick an orthonormal basis $(e_1, \dots, e_r)$ for U , which we can do by the Gram-Schmidt Orthogonalization Process 7.18. Let $v \in V$ be arbitrary. Define

$$P_U(v) := \langle v, e_1 \rangle e_1 + \dots + \langle v, e_r \rangle e_r \in U.$$

We have that

$$v = \underbrace{P_U(v)}_{\in U} + (v - P_U(v)). \quad (7.1)$$

We claim that $v - P_U(v) \in U^\perp$. Indeed,

$$\langle v - P_U(v), e_i \rangle = \langle v, e_i \rangle - \sum_{k=1}^r \langle v, e_k \rangle \cdot \langle e_k, e_i \rangle = \langle v, e_i \rangle - \langle v, e_i \rangle = 0.$$

This holds for every $1 \leq i \leq r$, hence

$$v - P_U(v) \perp \text{Sp}(e_1, \dots, e_r) = U,$$

i.e., $v - P_U(v) \in U^\perp$. Going back to Equation (7.1) we see that

$$v = \underbrace{P_U(v)}_{\in U} + \underbrace{(v - P_U(v))}_{\in U^\perp}.$$

Since $v \in V$ was arbitrary, we have that $U + U^\perp = V$. □

Remark 7.24. If $U \subseteq V$ is a finite dimensional subspace, then since $U^\perp$ is a complement of U in V , we have a canonical isomorphism $V \cong U \oplus U^\perp$.

Definition 7.25: The Orthogonal Projection onto a Subspace

Let V be an inner product space and $U \subseteq V$ a finite dimensional subspace. We have seen that $U^\perp$ is a complement of U in V ($\implies \forall v \in V, \exists! u \in U, w \in U^\perp$ s.t. $v = u + w$).

Define a map $P_U : V \rightarrow U$ as follows:

$$\forall v \in V, \exists! u \in U, w \in U^\perp \text{ s.t. } v = u + w.$$

We define

$$P_U(v) := u.$$

P_U is called **the orthogonal projection onto U** .

Proposition 7.26: Properties of P_U

P_U has the following properties:

1. P_U is a linear map.
2. $\text{Im } P_U = U, \text{ ker } P_U = U^\perp$.
3. $\forall v \in V : v - P_U(v) \in U^\perp$, i.e., $P_U + P_{U^\perp} = \text{id}_V$.
4. $\forall u \in U : P_U(u) = u$.
5. If $e_1, \dots, e_r$ is any orthonormal basis for U , then

$$P_U(v) = \sum_{i=1}^r \langle v, e_i \rangle e_i.$$

Corollary 7.27: The Complement of the Complement of a Subspace

Let V be an inner product space and $U \subseteq V$ a finite dimensional subspace. Then

$$(U^\perp)^\perp = U.$$

Corollary 7.28: Dimension of a Subspace and its Complement

If V is a finite dimensional inner product space and $U \subseteq V$ is a subspace, then

$$\dim U + \dim U^\perp = \dim V.$$

7.4 Orthogonal & Unitary Matrices

Definition 7.29: Orthogonal & Unitary Matrices

1. A matrix $A \in M_{n \times n}(\mathbb{R})$ is called **orthogonal** if its columns form an orthonormal basis in $\mathbb{R}^n$, w.r.t. the standard scalar product of $\mathbb{R}^n$. We denote the set of orthogonal $n \times n$ matrices by $O(n) \subseteq M_{n \times n}(\mathbb{R})$.
2. A matrix $A \in M_{n \times n}(\mathbb{C})$ is called **unitary** if its columns form an orthonormal basis in $\mathbb{C}^n$, w.r.t. the standard hermitian product on $\mathbb{C}^n$. We denote the set of all unitary $n \times n$ matrices by $U(n) \subseteq M_{n \times n}(\mathbb{C})$.

Lemma 7.30: Equivalent Conditions for Orthogonal/Unitary Matrices

- $A \in M_{n \times n}(\mathbb{R})$ is orthogonal $\iff A^T A = I_n \iff A A^T = I_n \iff A^{-1} = A^T$.
- $A \in M_{n \times n}(\mathbb{C})$ is unitary $\iff A^* A = I_n \iff A A^* = I_n \iff A^{-1} = A^*$.

Corollary 7.31

1. Let $A \in M_{n \times n}(\mathbb{R})$. Then $A \in O(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{R}^n$.
2. Let $A \in M_{n \times n}(\mathbb{C})$. Then $A \in U(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{C}^n$.

7.5 QR-Decomposition

Theorem 7.32: QR-Decomposition

1. Let $A \in GL_n(\mathbb{R})$. Then $\exists Q \in O(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{R})$ s.t.

$$A = QR.$$

2. Let $A \in GL_n(\mathbb{C})$. Then $\exists Q \in U(n)$ and an upper triangular matrix $R \in M_{n \times n}(\mathbb{C})$ s.t.

$$A = QR.$$

Proof. We'll prove 1 & 2 together.

Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

with $v_k \in K_{\text{col}}^n$, where $K = \mathbb{R}$ or $\mathbb{C}$. Since $A \in GL_n(K)$, $v_1, \dots, v_n$ form a basis for K_{col}^n . Apply the Gram-Schmidt orthogonalization process 7.18 to $v_1, \dots, v_n$ and obtain an orthonormal basis $e_1, \dots, e_n$ for K_{col}^n . Recall that $\forall 1 \leq j \leq n : \text{Sp}(v_1, \dots, v_j) = \text{Sp}(e_1, \dots, e_j)$. By Theorem 7.16 we also have that

$$v_j = \sum_{i=1}^j \langle v_j, e_i \rangle e_i \quad \forall 1 \leq j \leq n.$$

Therefore, we can write

$$\underbrace{\begin{pmatrix} | & & | & & | \\ v_1 & \dots & v_j & \dots & v_n \\ | & & | & & | \end{pmatrix}}_{=A} = \underbrace{\begin{pmatrix} | & & | & & | \\ e_1 & \dots & e_j & \dots & e_n \\ | & & | & & | \end{pmatrix}}_{=Q} \cdot \underbrace{\begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \dots & \langle v_j, e_1 \rangle & \dots & \langle v_n, e_1 \rangle \\ 0 & \langle v_2, e_2 \rangle & \dots & \langle v_j, e_2 \rangle & \dots & \langle v_n, e_2 \rangle \\ \vdots & \ddots & \ddots & \vdots & & \vdots \\ \vdots & & \ddots & \langle v_j, e_j \rangle & \dots & \langle v_n, e_j \rangle \\ \vdots & & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 & \langle v_n, e_n \rangle \end{pmatrix}}_{=R},$$

as desired. $\square$

Extension of QR-Decomposition for all matrices $A \in M_{n \times n}(K)$.

Theorem 7.33: Extension of QR-Decomposition

Let $A \in M_{n \times n}(K)$, where $K = \mathbb{R}$ or $\mathbb{C}$ and let $r := \text{rank}(A)$. Then $\exists Q \in O(n)/U(n)$ and

$$R = \begin{pmatrix} C & * \\ 0 & 0 \end{pmatrix} \in M_{n \times n}(K),$$

where $C \in M_{r \times r}(K)$ is upper triangular, s.t.

$$A = QR.$$

8 Dual Spaces & Inner Products

Let V be a vector space over K (any field). Recall that we have the dual space $V^* := \text{Hom}(V, K) =$ the space of linear maps $\varphi : V \rightarrow K$ (functionals). V^* is also a vector space over K .

Consider now an inner product space $(V, \langle \cdot, \cdot \rangle)$ over $K = \mathbb{R}$ or $\mathbb{C}$. Let $u \in V$. Define

$$\varphi_u : V \rightarrow K, \quad V \ni v \mapsto \langle v, u \rangle \in K.$$

φ_u is linear since, v is the first argument in the scalar/hermitian product.

Theorem 8.1: Canonical 'Almost' Isomorphism between V and V^*

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $\varphi \in V^*$. Then there exists a **unique** $u \in V$ s.t. $\varphi = \varphi_u$, i.e., $\varphi(v) = \varphi_u(v) \quad \forall v \in V$.

In fact, for $K = \mathbb{R}$ the map $\Phi : V \rightarrow V^*$, given by $\Phi(u) := \varphi_u$ is an isomorphism.

For $K = \mathbb{C}$, this map is injective and surjective but only $\mathbb{C}$ -anti linear, i.e.,

$$\begin{aligned} \Phi(u_1 + u_2) &= \Phi(u_1) + \Phi(u_2) & \forall u_1, u_2 \in V \\ \Phi(cu) &= \bar{c} \cdot \Phi(u) & \forall c \in \mathbb{C}, u \in V. \end{aligned}$$

So, at least for $K = \mathbb{R}$, a scalar product on V gives us a canonical isomorphism $V \cong V^*$. (But this isomorphism depends on the scalar product!)

8.1 The Adjoint Map

(Everything in this Subsection should be learned by heart.) Let V, W be vector spaces over K . Let $T : V \rightarrow W$ be a linear map. We have seen in Subsection 4.2 that T induces a linear map (the dual map)

$$T^* : W^* \rightarrow V^*, \quad T^*(\varphi) := \varphi \circ T.$$

Assume now that $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ are inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. Assume also that $\dim V < \infty$. Given T as above, we can define $T' : W \rightarrow V$ using our 'canonical' identifications $W \cong W^*$, $V \cong V^*$. (No problem for $K = \mathbb{R}$, but we have to be careful with $K = \mathbb{C}$.)

$$\begin{array}{ccc} W^* & \xrightarrow{T^*} & V^* \\ \uparrow \Phi_W & & \downarrow \Phi_V^{-1} \\ W & \xrightarrow{\quad T' \quad} & V \end{array}$$

i.e. $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$. We've also used that $\dim V < \infty$, b.c. then Φ_V is bijective.

What properties does T' have? Note that T' is uniquely determined by

$$\begin{aligned} \Phi_V \circ T' &= T^* \circ \Phi_W &\iff \forall w \in W : \Phi_V \circ T'(w) &= T^* \circ \Phi_W(w) \\ &\iff \varphi_{T'(w)} &= T^* \circ \varphi_w \quad (\in V^*) \\ &\iff \forall v \in V : \varphi_{T'(w)}(v) &= (T^* \circ \varphi_w)(v) \\ &\iff \langle v, T'(w) \rangle_V &= \varphi_w(T(v)) = \langle T(v), w \rangle_W. \end{aligned}$$

In other words,

$$\langle Tv, w \rangle_W = \langle v, T'w \rangle_V \quad \forall v \in V, w \in W.$$

Claim: T' is a linear map.

Proof. For $K = \mathbb{R}$, Φ_W, Φ_V and T^* are linear. Therefore, T' is linear too.

For $K = \mathbb{C}$, Φ_W, Φ_V are additive but only $\mathbb{C}$ -anti linear. Therefore, Φ_V^{-1} is also additive and $\mathbb{C}$ -anti linear. Indeed, let $y_1, y_2 \in V^*$. Since Φ_V is bijective, we can find unique $x_1, x_2 \in V$ s.t. $\Phi_V(x_1) = y_1$ and $\Phi_V(x_2) = y_2$. Since Φ_V is additive we have that

$$\Phi_V^{-1}(y_1 + y_2) = \Phi_V^{-1}(\Phi_V(x_1) + \Phi_V(x_2)) = \Phi_V^{-1}(\Phi_V(x_1 + x_2)) = x_1 + x_2 = \Phi_V^{-1}(y_1) + \Phi_V^{-1}(y_2).$$

Similarly, let $c \in \mathbb{C}, y \in V^*$. We again can find a unique $x \in V$ s.t. $\Phi_V(x) = y$. Since Φ_V is $\mathbb{C}$ -anti linear we have that

$$\Phi_V^{-1}(cy) = \Phi_V^{-1}(c \cdot \Phi_V(x)) = \Phi_V^{-1}(\Phi_V(\bar{c}x)) = \bar{c} \cdot x = \bar{c} \cdot \Phi_V^{-1}(y).$$

So, we have shown that Φ_V^{-1} is additive and $\mathbb{C}$ -anti linear as well.

Now, we return to the proof of the claim. We recall that T^* is linear. Let $w_1, w_2 \in W$ be arbitrary. Since T^* is in particular additive and $T' = \Phi_V^{-1} \circ T^* \circ \Phi_W$, we can conclude that T' is also additive.

Now, let $c \in \mathbb{C}, w \in W$. Since T^* is linear and Φ_V^{-1}, Φ_W are $\mathbb{C}$ -anti linear we have that

$$\begin{aligned} T'(cw) &= \Phi_V^{-1} \circ T^* \circ \Phi_W(cw) = \Phi_V^{-1} \left(T^*(\bar{c} \cdot \Phi_W(w)) \right) = \Phi_V^{-1} \left(\bar{c} \cdot T^*(\Phi_W(w)) \right) \\ &= c \cdot \Phi_V^{-1} \left(T^*(\Phi_W(w)) \right) = c \cdot T'(w). \end{aligned}$$

This proves the claim. $\square$

Notation: The linear map $T' : W \rightarrow V$ is denoted many times by $T^* : W \rightarrow V$. This makes sense b.c. of the 'canonical' identifications. But we have to assume additionally that $\dim W < \infty$. The map $T^* : W \rightarrow V$ is called the **adjoint** of T .

8.2 Basic Properties of Adjoint Maps

Lemma 8.2: Properties of Adjoint Maps

Let V, W, U be finite dimensional inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. Let $S, T : V \rightarrow W$ and $R : W \rightarrow U$ be linear maps. Then:

1. T^* is a linear map.
2. $(S + T)^* = S^* + T^*$.
3. $\forall \lambda \in K : (\lambda T)^* = \bar{\lambda} T^*$.
4. $(T^*)^* = T$.
5. $(id_V)^* = id_V$.
6. $(R \circ T)^* = T^* \circ R^*$.

Proof. Let $y_1, y_2 \in V$. We'll repeatedly use that if $\forall x \in V : \langle x, y_1 \rangle = \langle x, y_2 \rangle$, then $y_1 = y_2$. This is because

$$\begin{aligned} \forall x \in V : \langle x, y_1 \rangle &= \langle x, y_2 \rangle &\implies & \forall x \in V : \langle x, y_1 - y_2 \rangle = 0 \\ &\implies y_1 - y_2 = 0 &\implies & y_1 = y_2, \end{aligned}$$

where we've used 2. of Lemma 7.3

- 1) Has already been proven in Subsection 8.1.
- 2) Let $w \in W, v \in V$. By the defining properties of T^*, S^* we have that

$$\langle v, (S + T)^* w \rangle = \langle (S + T)v, w \rangle = \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^* w \rangle + \langle v, T^* w \rangle = \langle v, (S^* + T^*)w \rangle.$$

This holds for every $v \in V$. Therefore, we have that

$$(S + T)^* w = (S^* + T^*)w.$$

Since, this holds for every $w \in W$, we conclude that $(S + T)^* = (S^* + T^*)$.

- 3) Let $\lambda \in K, v \in V$ and $w \in W$. Then we have that

$$\begin{aligned} \langle v, (\lambda T)^* w \rangle &= \langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^* w \rangle \\ &= \langle v, \bar{\lambda} T^* w \rangle. \end{aligned}$$

Since this holds for every $v \in V$, we get that

$$(\lambda T)^* w = \bar{\lambda} T^* w.$$

Since this holds for every $w \in W$, we conclude that $(\lambda T)^* = \bar{\lambda} T^*$.

4) Let $v \in V$, $w \in W$. Then we have that

$$\langle w, (T^*)^* v \rangle = \langle T^* w, v \rangle = \overline{\langle v, T^* w \rangle} = \overline{\langle T v, w \rangle} = \langle w, T v \rangle.$$

This holds for every $w \in W$. Therefore, we have that $(T^*)^* v = T v$. And since this holds for every $v \in V$, we get that $(T^*)^* = T$.

5) Let $v \in V$. Then we have that

$$\langle v, (id_V)^* v \rangle = \langle id_V v, v \rangle = \langle v, v \rangle = \langle v, id_V v \rangle.$$

Since this holds for every $v \in V$ we conclude that

$$(id_V)^* v = id_V v \quad \implies \quad (id_V)^* = id_V.$$

6) Let $v \in V$, $u \in U$. Then we have that

$$\begin{aligned} \langle v, (R \circ T)^* u \rangle &= \langle (R \circ T) v, u \rangle = \langle R(T(v)), u \rangle = \langle T v, R^* u \rangle \\ &= \langle v, T^* (R^* (u)) \rangle = \langle v, T^* \circ R^* u \rangle. \end{aligned}$$

This holds for every $v \in V$. Therefore,

$$(R \circ T)^* u = T^* \circ R^* u.$$

Since this holds for every $u \in U$, we conclude that $(R \circ T)^* = T^* \circ R^*$. □

Lemma 8.3: Kernel & Image of Adjoint Maps

Let V, W be finite dimensional inner product spaces. Let $T : V \rightarrow W$ and $T^* : W \rightarrow V$ be the adjoint of T . Then

1. $\ker T^* = (\operatorname{Im} T)^\perp$.
2. $\operatorname{Im} T^* = (\ker T)^\perp$.
3. $\ker T = (\operatorname{Im} T^*)^\perp$.
4. $\operatorname{Im} T = (\ker T^*)^\perp$.

8.3 Matrix Representation of Adjoint Maps

Proposition 8.4: Matrix Representation of Adjoint Maps

Let V, W be finite dimensional inner product spaces and $TV \rightarrow W$ a linear map. Let $\mathcal{B} = (e_1, \dots, e_n)$ be an orthonormal basis for V and $\mathcal{C} = (f_1, \dots, f_m)$ an orthonormal basis for W . Then

$$[T^*]_{\mathcal{B}}^{\mathcal{C}} = \left(\overline{[T]_{\mathcal{C}}^{\mathcal{B}}} \right)^T = \left([T]_{\mathcal{C}}^{\mathcal{B}} \right)^*,$$

where the last asterisk denotes the adjoint matrix.

Corollary 8.5

Let $A \in M_{m \times n}(K)$, where $K = \mathbb{R}$ or $\mathbb{C}$ and let $T_A : K^n \rightarrow K^m$ be the linear map $T_A(v) = Av$. Endow K^n and K^m with their standard inner product. Then

$$(T_A)^* = T_{A^*}.$$

Proposition 8.6: Properties of the Adjoint Matrix

Let $K = \mathbb{R}$ or $\mathbb{C}$, $A, B \in M_{n \times m}(K)$, $C \in M_{m \times p}(K)$. Then:

1. $(A + B)^* = A^* + B^*$.
2. $(\lambda A)^* = \bar{\lambda} A^* \quad \forall \lambda \in K$.
3. $(A^*)^* = A$.
4. $I_n^* = I_n$.
5. $(AC)^* = C^* A^*$.

9 Spectral Theorems

Definition 9.1: Orthogonal Diagonalizability

Let V be an inner product space and $T : V \rightarrow V$ an endomorphism. We say that T is **orthogonally diagonalizable** if there exists an orthonormal basis for V consisting of eigenvectors of T .

Lemma 9.2: Consequence of Orthogonal Diagonalizability

If T is orthogonally diagonalizable, then

$$T \circ T^* = T^* \circ T.$$

Definition 9.3: Normal Endomorphism/Matrix

Let V be an inner product space. An endomorphism $T : V \rightarrow V$ is called **normal** if

$$T \circ T^* = T^* \circ T.$$

A matrix $A \in M_{n \times n}(K)$ ($K = \mathbb{R}, \mathbb{C}$) is called **normal** if

$$AA^* = A^*A.$$

This is equivalent to saying that $T_A : K^n \rightarrow K^n$ is normal, if we endow K^n with its standard inner product.

We've seen in Lemma 9.2 that if $T : V \rightarrow V$ is orthogonally diagonalizable, then T is normal. Is the converse statement also true?

Over $K = \mathbb{R}$ the answer is **no!**, e.g.,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

$T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a clockwise rotation by 90° , so there exist no eigenvalues in $\mathbb{R}$. But $AA^* = A^*A = I$.

Lemma 9.4: Properties of a Normal Endomorphism

Let V be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $T : V \rightarrow V$ be a normal endomorphism. Then:

1. $\forall v \in V : \|Tv\| = \|T^*v\|$.
2. $\forall \lambda \in K$ the endomorphism $T - \lambda \cdot id_V$ is also normal.
3. If v is an eigenvector of T with eigenvalue λ , then v is also an eigenvector of T^* with eigenvalue $\bar{\lambda}$.
4. Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively. If $\lambda_1 \neq \lambda_2$, then $\langle v_1, v_2 \rangle = 0$.

Proof. **1)** Let $v \in V$. Then by the properties of the adjoint map T^* and since T is normal, we have that

$$\begin{aligned} \|Tv\|^2 &= \langle Tv, Tv \rangle = \langle v, T^* \circ Tv \rangle = \langle v, T \circ T^*v \rangle \\ &= \overline{\langle T \circ T^*v, v \rangle} = \overline{\langle T^*v, T^*v \rangle} = \langle T^*v, T^*v \rangle = \|T^*v\|^2. \end{aligned}$$

Since the norm is non-negative, we conclude that

$$\|Tv\| = \|T^*v\|.$$

2) Let $\lambda \in K$. By properties of adjoint maps we have that

$$(T - \lambda \cdot id_V) \circ (T - \lambda \cdot id_V)^* = (T - \lambda \cdot id_V) \circ (T^* - \bar{\lambda} \cdot id_V) = T \circ T^* - \bar{\lambda} \cdot T - \lambda \cdot T^* + \lambda \bar{\lambda} \cdot id_V, \quad (9.1)$$

and

$$(T - \lambda \cdot id_V)^* \circ (T - \lambda \cdot id_V) = (T^* - \bar{\lambda} \cdot id_V) \circ (T - \lambda \cdot id_V) = T^* \circ T - \lambda \cdot T^* - \bar{\lambda} \cdot T + \bar{\lambda} \lambda \cdot id_V. \quad (9.2)$$

By the normality of T we get that Equation (9.1) = (9.2), so $(T - \lambda \cdot id_V)$ is normal.

3) Let v be an eigenvector of T with eigenvalue λ . Then $(T - \lambda \cdot id_V)v = 0$. By **2)**, $(T - \lambda \cdot id_V)$ is normal. Hence by **1)** we have

$$\begin{aligned} \|(T^* - \bar{\lambda} \cdot id_V)v\| &= \|(T - \lambda \cdot id_V)^*v\| = \|(T - \lambda \cdot id_V)v\| = 0 \\ \implies (T^* - \bar{\lambda} \cdot id_V)v &= 0. \end{aligned}$$

This proves that v is an eigenvector of T^* with eigenvalue $\bar{\lambda}$.

4) Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively, where $\lambda_1 \neq \lambda_2$. Then, by properties of adjoint maps and **3)**, we have that

$$\begin{aligned} \lambda_1 \langle v_1, v_2 \rangle &= \langle \lambda_1 v_1, v_2 \rangle = \langle T v_1, v_2 \rangle = \langle v_1, T^* v_2 \rangle = \langle v_1, \bar{\lambda}_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle \\ \implies (\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle &= 0. \end{aligned}$$

Since $\lambda_1 \neq \lambda_2$, we conclude that $\langle v_1, v_2 \rangle = 0$. □

Theorem 9.5: Spectral Theorem over $\mathbb{C}$

Let V be a finite dimensional inner product space over $\mathbb{C}$ and $T : V \rightarrow V$ a linear map. Then T is orthogonally diagonalizable if and only if T is normal.

Proof. $\implies$: Has already been proven in Lemma 9.2

$\impliedby$: By the fundamental Theorem of Algebra 6.26, T is trigonalizable, i.e., there exists a basis $\mathcal{C} = (u_1, \dots, u_n)$ for V s.t. $[T]_{\mathcal{C}}^{\mathcal{C}}$ is an upper triangular matrix. By applying the Gram-Schmidt orthogonalization process 7.18 to the basis $\mathcal{C}$ we obtain a new basis $\mathcal{B} = (v_1, \dots, v_n)$ which is orthonormal and s.t. $[T]_{\mathcal{B}}^{\mathcal{B}}$ is still upper triangular, b.c. $\text{Sp}(u_1, \dots, u_j) = \text{Sp}(v_1, \dots, v_j)$ for every $1 \leq j \leq n$.

We claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is actually diagonal. To see this write $[T]_{\mathcal{B}}^{\mathcal{B}} = (a_{ij})$. Since $\mathcal{B}$ is orthonormal, we have $\|T v_1\|^2 = |a_{11}|^2$. Since $[T^*]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^*$ we also have

$$\begin{aligned} [T^*]_{\mathcal{B}}^{\mathcal{B}} &= \begin{pmatrix} \bar{a}_{11} & 0 & \dots & 0 \\ \bar{a}_{12} & \bar{a}_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ \bar{a}_{1n} & \bar{a}_{2n} & \dots & \bar{a}_{nn} \end{pmatrix} \\ \implies \|T^* v_1\|^2 &= \sum_{k=1}^n |\bar{a}_{1k}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2. \end{aligned}$$

As T is normal, we have that $\|T v_1\|^2 = \|T^* v_1\|^2$, hence $a_{12} = \dots = a_{1n} = 0$. This shows that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \bar{a}_{11} & 0 & \dots & 0 \\ 0 & * & \dots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

By the same argumentation, we have that $\|T v_2\|^2 = |a_{22}|^2$ and $\|T^* v_2\|^2 = |a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2$.

As $\|Tv_2\| = \|T^*v_2\|$ we obtain $a_{23} = \dots = a_{2n} = 0$.

We can continue the proof by induction. Suppose the claim holds for k . So, we have that $\|Tv_{k+1}\|^2 = |a_{k+1,k+1}|^2$, as $a_{l,k+1} = 0 \quad \forall l < k+1$. Since $[T^*]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^*$, we have that

$$\|T^*v_{k+1}\|^2 = |a_{k+1,k+1}|^2 + \dots + |a_{k+1,n}|^2.$$

Since T is normal, we get that $\|T^*v_{k+1}\| = \|Tv_{k+1}\|$. Therefore, we conclude that

$$|a_{k+1,k+2}| = \dots = |a_{k+1,n}| = 0.$$

This proves the claim. □

Remark 9.6. In the proof above we used $K = \mathbb{C}$ only, when we said that T is trigonalizable. Therefore, we arrive at the following corollary.

Corollary 9.7

Let V be a finite dimensional inner product space over $\mathbb{R}$ and $T : V \rightarrow V$ an endomorphism, which is normal. If T is trigonalizable, then T is orthogonally diagonalizable.

We've seen that over $\mathbb{R}$ $T \in \text{End}(V)$ is normal $\not\Rightarrow T$ is orthogonally diagonalizable.

Definition 9.8: Self Adjoint

Let V be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$, and $T \in \text{End}(V)$. T is called **self adjoint** if $T^* = T$.

A matrix $A \in M_{n \times n}(K)$ is called **self adjoint** if $A^* = A$. (Over $\mathbb{R}$ a self adjoint matrix is a symmetric matrix.)

Lemma 9.9: A Condition for Self Adjoint

Let V be a finite dimensional inner product space over $\mathbb{R}$ and $T \in \text{End}(V)$ which is orthogonally diagonalizable. Then T is self adjoint.

Lemma 9.10

Let V be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Then $T \in \text{End}(V)$ is self adjoint if and only if for every orthonormal basis $\mathcal{B}$ of V , the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint.

Theorem 9.11: Spectral Theorem over $\mathbb{R}$

Let V be an Euclidean space(, i.e., inner product space over $\mathbb{R}$) of finite dimension. Then T is orthogonally diagonalizable if and only if T is self adjoint.

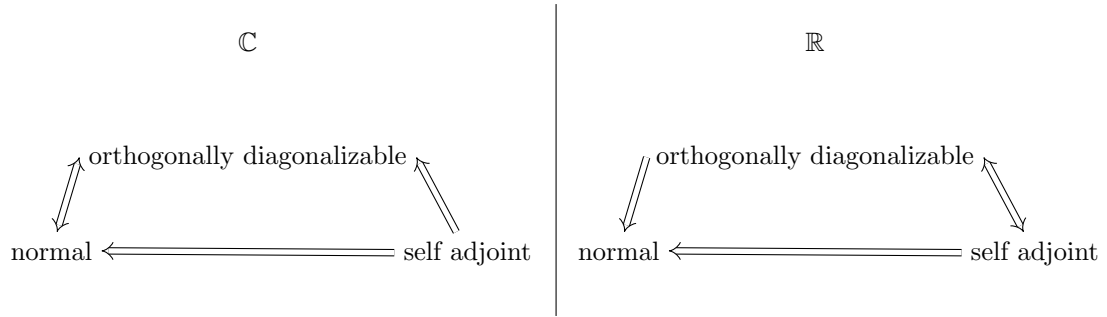
For the proof of the above Theorem the following Lemma is needed.

Lemma 9.12: Properties of Self Adjoint Maps

Let V be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$ and $T \in \text{End}(V)$ be self adjoint. Then

1. All eigenvalues of T are real.
2. p_T factors into a product of linear factors over K .

In the Figure below we illustrated the logical relations regarding the Spectral theorem.



Theorem 9.13: Spectral Theorem for Matrices

1. If $A \in M_{n \times n}(\mathbb{C})$ is normal, then $\exists U \in U(n)$ s.t.

$$\underbrace{U^{-1}}_{= U^*} A U$$

is diagonal.

2. If $A \in M_{n \times n}(\mathbb{R})$ is a symmetric matrix, then $\exists O \in O(n)$ s.t.

$$\underbrace{O^{-1}}_{= O^T} A O$$

is diagonal.

10 Orthogonal & Unitary maps / Isometries

Definition 10.1: Linear Isometry

Let V, W be inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. A map $T : V \rightarrow W$ is called a **linear isometry** if T is linear and

$$\forall v_1, v_2 \in V : \langle T v_1, T v_2 \rangle_W = \langle v_1, v_2 \rangle_V,$$

i.e., T preserves the inner product.

Remark 10.2. If $T : V \rightarrow W$ is a linear isometry, then:

1. $\forall v \in V : \|T v\| = \|v\|.$

2. $\forall p, q \in V : d(Tp, Tq) = d(p, q)$, where $d(p, q) := \|p - q\|$.

3. T is injective. Indeed, if $v \in \ker T$, then

$$0 = \|Tv\| = \|v\| \implies v = 0.$$

So, $\ker T = \{0\}$.

From now on we'll concentrate on the case $W = V$, i.e., $T \in \text{End}(V)$. If such a map T is an isometry we also call it **orthogonal endomorphism** if $K = \mathbb{R}$, or **unitary endomorphism** if $K = \mathbb{C}$.

Theorem 10.3: Equivalent Statements to Isometry

Let V be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$, and $T \in \text{End}(V)$. The following are equivalent:

1. T is an isometry.
2. $\forall v \in V : \|Tv\| = \|v\|$.
3. For every orthonormal system $e_1, \dots, e_k$ of V , the list of vectors $Te_1, \dots, Te_k$ is also an orthonormal system.
4. There exists an orthonormal basis $e_1, \dots, e_n$ of V s.t. $Te_1, \dots, Te_n$ is also an orthonormal basis.
5. $TT^* = id_V$.
6. $T^*T = id_V$.
7. $T^* = T^{-1}$.
8. T^* is an isometry.

Back to matrices.

Proposition 10.4: Equivalent Condition for a Orthogonal/Unitary Map

Let V be an n -dimensional inner product space over $K = \mathbb{R}$ (respectively $\mathbb{C}$). Let $\mathcal{B}$ be an orthonormal basis for V . Let $T \in \text{End}(V)$. Then T is orthogonal (respectively unitary) if and only if $[T]_{\mathcal{B}}^{\mathcal{B}} \in O(n)$ (respectively $[T]_{\mathcal{B}}^{\mathcal{B}} \in U(n)$).

Note that $\forall A \in O(n)$ we have $\det A = \pm 1$. Indeed,

$$1 = \det I = \det(AA^T) = \det A \cdot \det A.$$

Similarly, $\forall A \in U(n)$, $|\det A| = 1$. Indeed,

$$1 = \det(AA^*) = \det A \cdot \det \overline{A}^T = \det A \cdot \det \overline{A} = \det A \cdot \overline{\det A} = |\det A|^2.$$

Definition 10.5: Special Orthogonal/Unitary Group

The set

$$SO(n) := \{A \in O(n) \mid \det A = 1\} \subseteq O(n) \subseteq GL_n(\mathbb{R})$$

is called the **special orthogonal group**.

The set

$$SU(n) := \{A \in U(n) \mid \det A = 1\} \subseteq U(n) \subseteq GL_n(\mathbb{C})$$

is called the **special unitary group**.

Proposition 10.6: Group Structure of $O(n)/SO(n)/U(n)/SU(n)$

Let $G = O(n), SO(n), U(n), SU(n)$. Then:

1. $I_n \in G$.
2. $\forall A, B \in G : A \cdot B \in G$.
3. $\forall A \in G : A^{-1} \in G$.

10.1 Classification of the Elements of $O(2), O(3)$ etc.**Lemma 10.7: Eigenvalue of an Orthogonal Matrix**

Let $A \in O(n)$. If λ is an eigenvalue of A , then $\lambda = \pm 1$.

Proposition 10.8: Classification of the Elements of $O(2)$

Let $A \in O(2)$.

1. If $\det A = 1$, then $\exists \theta \in [0, 2\pi)$ s.t.

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} := R_\theta.$$

The map $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is an anti-clockwise rotation by the angle θ .

2. If $\det A = -1$, then there exists a basis $\mathcal{B} = (v_1, v_2)$ for $\mathbb{R}^2$ s.t. $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ has the following representation in the basis $\mathcal{B}$

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (10.1)$$

In other words T_A performs a reflection w.r.t. the line $l = \text{Sp}(v_1)$. Moreover, $\exists \theta \in [0, 2\pi)$ s.t.

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The basis $\mathcal{B}$ for which Equation (10.1) holds, is given by

$$v_1 = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix}, v_2 = \begin{pmatrix} \cos(\frac{\theta}{2} + \frac{\pi}{2}) \\ \sin(\frac{\theta}{2} + \frac{\pi}{2}) \end{pmatrix}.$$

Summary:

$SO(2) =$ rotations

$O(2) \setminus SO(2) =$ reflections.

Corollary 10.9

Let V be a 2-dimensional Euclidean space and $T : V \rightarrow V$ a linear isometry. Then $\det T = \pm 1$. Moreover, there exists an orthonormal basis $\mathcal{B}$ for V s.t.

1. If $\det T = 1$, then $[T]_{\mathcal{B}}^{\mathcal{B}} = R_\theta$.

2. If $\det T = -1$, then

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Proposition 10.10: Classification of the Elements of $SO(3)$

Let $A \in SO(3)$. Then 1 is an eigenvalue of A . Moreover, there exists an orthonormal basis $\mathcal{B} = (v_1, v_2, v_3)$ and $\theta \in [0, 2\pi)$ s.t.

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & & R_{\theta} \end{array} \right).$$

So, T_A is a rotation with angle θ around the axis $\text{Sp}(v_1)$. Moreover,

(a) If -1 is an eigenvalue of A , then $\theta = \pi$, hence

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(b) If -1 is **not** an eigenvalue of A ($\iff 1$ is the only eigenvalue), then $\theta \neq \pi$.

Proof. The characteristic polynomial $p_A(x)$ of A is a polynomial of degree 3 and with real coefficients. By the intermediate value theorem (Analysis I) there exists a real zero $\lambda \in \mathbb{R}$ (i.e., $p_A(\lambda) = 0$). Therefore, A has at least one real eigenvalue. By Lemma 10.7, $\lambda = \pm 1$. Let $v_1 \in \mathbb{R}^3 \setminus \{0\}$ be an eigenvector of λ . We can normalize v_1 and still have an eigenvector for λ . So, let's assume $\|v_1\| = 1$. Consider the subspace

$$L := v_1^\perp = \{v \in \mathbb{R}^3 \mid \langle v, v_1 \rangle = 0\}.$$

Since A is orthogonal, we have that

$$\forall v \in L : Av \in L,$$

i.e., $T_A(L) \subseteq L$, i.e., L is invariant under T_A . Indeed, if $v \in L$, then since $A \in O(3)$

$$\begin{aligned} 0 &= \langle v, v_1 \rangle = \langle Av, Av_1 \rangle = \langle Av, \pm v_1 \rangle = \pm \langle Av, v_1 \rangle \\ \implies Av &\in L. \end{aligned}$$

We also have $\mathbb{R}^3 = \text{Sp}(v_1) \oplus L$, hence $\dim L = 2$.

Now $T_A|_L : L \rightarrow L$ is a linear isometry. Let $\mathcal{L} = (w_1, w_2)$ be an orthonormal basis for L . By Proposition 10.4, we have that

$$Q := [T_A|_L]_{\mathcal{L}}^{\mathcal{L}} \in O(2).$$

Case (a): $\lambda = 1$. Take $\mathcal{B} = (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & & Q \end{array} \right).$$

Now, since $A \in SO(3)$, we have

$$\begin{aligned} 1 &= \det A = \det(T_A) = 1 \cdot \det Q \\ \implies \det Q &= 1. \end{aligned}$$

So, $Q \in SO(2)$. By Proposition 10.10, $\exists \theta \in [0, 2\pi)$ s.t. $Q = R_\theta$. Therefore,

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & R_\theta & \end{array} \right).$$

Case (b): $\lambda = -1$. By the same argument as for case (a), we get that

$$\begin{aligned} 1 &= \det A = (-1) \cdot \det(T_A|_L) \\ \implies \det(T_A|_L) &= -1. \end{aligned}$$

By Corollary ??, there exists an orthonormal basis $\mathcal{L} = (w_1, w_2)$ for L s.t.

$$[T_A|_L]_{\mathcal{L}}^{\mathcal{L}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Take $\mathcal{B}' = (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

And for $\mathcal{B} = (w_1, v_1, w_2)$ we have

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & R_\pi & \end{array} \right).$$

Note that in case (b), 1 is an eigenvalue of A as well. □

Proposition 10.11: Classification of $O(3) \setminus SO(3)$

Let $A \in O(3) \setminus SO(3)$. Then there exists an orthonormal basis $\mathcal{B} = (v_1, v_2, v_3)$ for $\mathbb{R}^3$ s.t.

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} -1 & 0 & 0 \\ \hline 0 & & \\ 0 & R_\theta & \end{array} \right)$$

for some $\theta \in [0, 2\pi)$. (So T_A does a rotations by θ around $\text{Sp}(v_1)$, followed by a reflection w.r.t. $\text{Sp}(v_2, v_3)$.)

11 Bilinear & Quadratic Forms

Definition 11.1: Bilinear Form

Let V be a vector space over K . A function $B : V \times V \rightarrow K$ is called a **bilinear form** if the following two conditions are satisfied:

1. $\forall w \in V$ the map $B(\cdot, w) : V \rightarrow K$, $v \mapsto B(v, w)$ is linear.
2. $\forall v \in V$ the map $B(v, \cdot) : V \rightarrow K$, $w \mapsto B(v, w)$ is linear.

Lemma 11.2: Matrix Representation of a Bilinear Form

Let V be a vector space over K and $\mathcal{C} = (e_1, \dots, e_n)$ a basis for V . Let $B : V \times V \rightarrow K$ be a bilinear form. Then $\exists [B]_{\mathcal{C}} \in M_{n \times n}(K)$ s.t.

$$B\left(\sum_{i=1}^n c_i e_i, \sum_{j=1}^n d_j e_j\right) = \begin{pmatrix} c_1 & \dots & c_n \end{pmatrix} \cdot [B]_{\mathcal{C}} \cdot \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} \quad \forall c_i, d_j \in K.$$

The matrix $[B]_{\mathcal{C}}$ is called the matrix representation of B in the basis $\mathcal{C}$.

We saw also that for $T \in \text{End}(V)$ and $\mathcal{C}$ a basis for V we have a matrix representation $[T]_{\mathcal{C}}^{\mathcal{C}} \in M_{n \times n}(K)$, and that if $\mathcal{D}$ is another basis for V , then

$$\begin{aligned} [T]_{\mathcal{D}}^{\mathcal{D}} &= \underbrace{[id_V]_{\mathcal{D}}^{\mathcal{C}}}_{([id_V]_{\mathcal{C}}^{\mathcal{D}})^{-1}} \cdot [T]_{\mathcal{C}}^{\mathcal{C}} \cdot [id_V]_{\mathcal{C}}^{\mathcal{D}}. \end{aligned} \quad (11.1)$$

(Corollary 3.17). What happens for bilinear forms?

Lemma 11.3: Change of Bases for a Bilinear Form

Let V be a finite dimensional vector space over K and $\mathcal{C}, \mathcal{D}$ be two bases for V . Let $B : V \times V \rightarrow K$ be a bilinear form. Then

$$[B]_{\mathcal{D}} = ([id_V]_{\mathcal{C}}^{\mathcal{D}})^T \cdot [B]_{\mathcal{C}} \cdot [id_V]_{\mathcal{C}}^{\mathcal{D}}.$$

(This is quite different from Equation (11.1)).

Definition 11.4: Symmetric/Positive Definite Bilinear Form & Quadratic Form

1. A bilinear form $B : V \times V \rightarrow K$ is called **symmetric** if

$$\forall v, w \in V : B(v, w) = B(w, v).$$

2. Assume that $K = \mathbb{R}$. A bilinear form $B : V \times V \rightarrow K$ is called **positive definite** if

$$\forall v \in V : B(v, v) > 0.$$

3. For a bilinear form $B : V \times V \rightarrow K$, we define

$$q_B : V \rightarrow K, \quad q_B(v) := B(v, v).$$

We call q_B the **quadratic form** associated to B . Sometimes we just denote it by q .

Remark 11.5. Note that $B \in M_{n \times n}(K)$ and $B^T \in M_{n \times n}(K)$ give the same quadratic form $q : K^n \rightarrow K$. Indeed, $\forall v \in K^n$

$$q_B(v) = v^T B v = (v^T B v)^T = v^T B^T v = q_{B^T}(v),$$

where we used that $q_B(v)$ is a scalar.

Assume now that $2 \neq 0$ in K , then (i.e., $1 + 1 \neq 0$). Then also $\frac{1}{2}(B + B^T)$ gives the same quadratic

form as B .

Conclusion: if $2 \neq 0$ in K , then for the discussion on quadratic forms $q : K^n \rightarrow K$, we can assume they all come from symmetric matrices / bilinear forms.

Proposition 11.6: Polarization Identity

Assume $2 \neq 0$ in K . (note that then also $4 \neq 0$.) If B is symmetric and defines $q := q_B$, then $\forall v, w \in V$ we have

1.

$$B(v, w) = \frac{1}{4}q(v + w) - \frac{1}{4}q(v - w).$$

2.

$$B(v, w) = \frac{1}{2}(q(v + w) - q(v) - q(w)).$$